Unit-2- part-I

IoT &M2M

- M2M & IoT-A basic Perspective-Introduction
- Differences and similarities between M2M & IoT
- SDN for IoT
- NFV for IoT
- M2M value Chains
- IoT value Chains
- An emerging Industrial Structure for IoT
- Information Driven Global Value Chain and
- Global Information monopolies

Unit-2- part-II

IoT Architecture

- **IoT Architecture Components**
- **Comparing IoT Architectures**
- **A Simplified IoT Architecture**
- **Core IoT functional Stack**
- **Iot data management and compute stack**

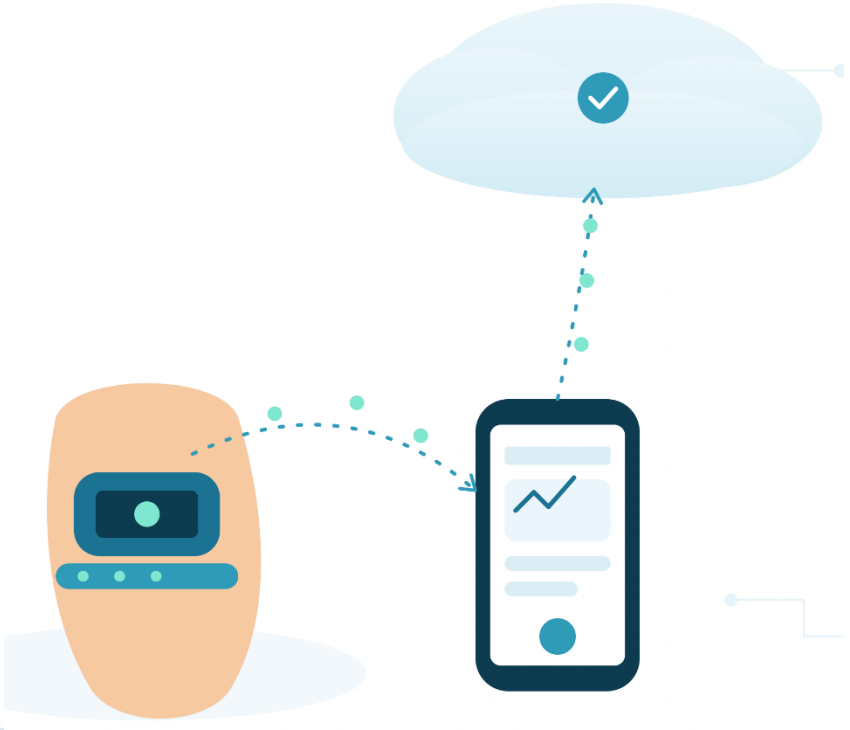
M2M & IoT-A basic Perspective-Introduction



M2M & IoT-A basic Perspective-Introduction

Machines conversing

Devices exchanging data directly, with little or no human intervention.



Data reaching the cloud

Information is collected, transmitted, and stored for processing.

Smart decisions

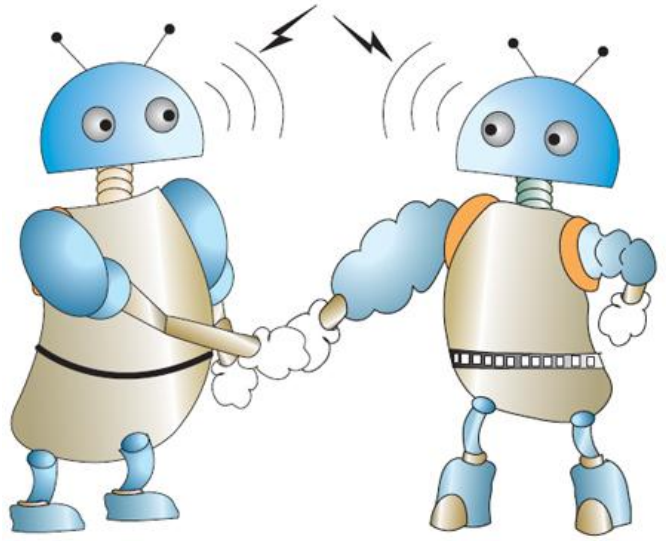
Systems sense, analyse, and act — automatically and in real time.

M2M & IoT-A basic Perspective-Introduction

What is IoT & M2M?

- **Provide Remote access** – IoT and M2M provide access to information without human intervention.
- **M2M** – M2M provides direct communication between individual machines or devices. It is designed to communicate between devices (Machines) for a specific purpose.





M2M & IoT-A basic Perspective-Introduction

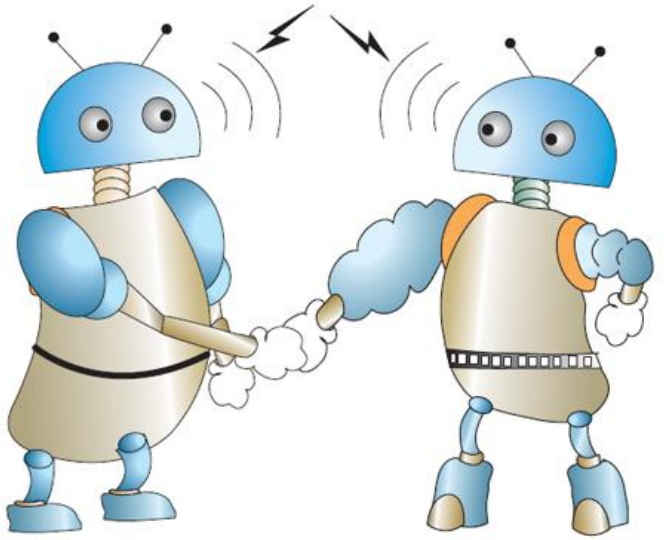
What is IoT & M2M?

- **IoT** – IoT is a broader concept for internet communication between devices. It involves a wide range of devices, sensors, actuators, and applications that communicate with the internet.

- M2M uses non-IP-based proprietary networks and IoT uses broad networks protocol based on IP.

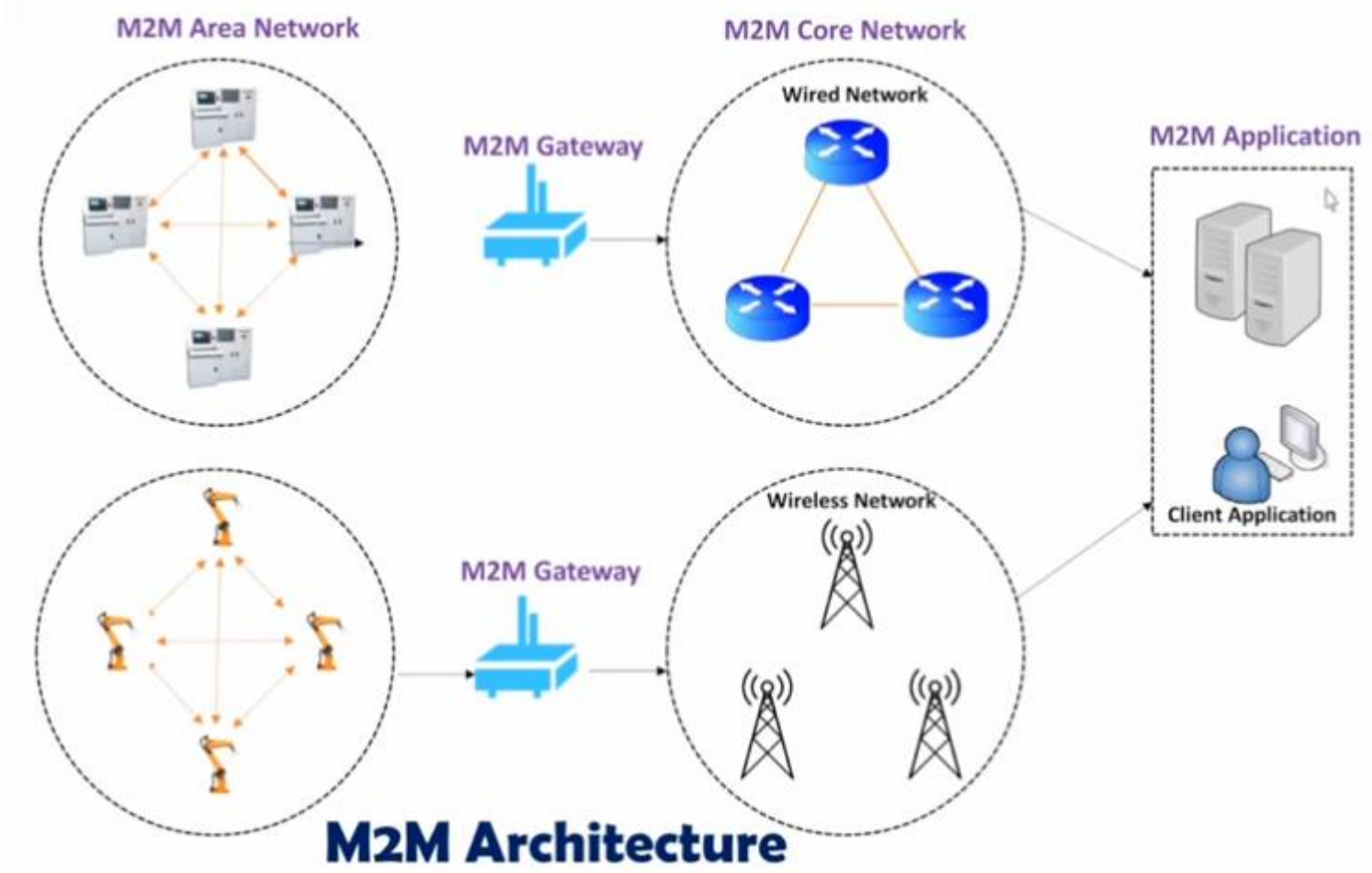
Example of Non-IP based networks:

Zigbee, Bluetooth, LoRaWAN…



M2M & IoT-A basic Perspective-Introduction



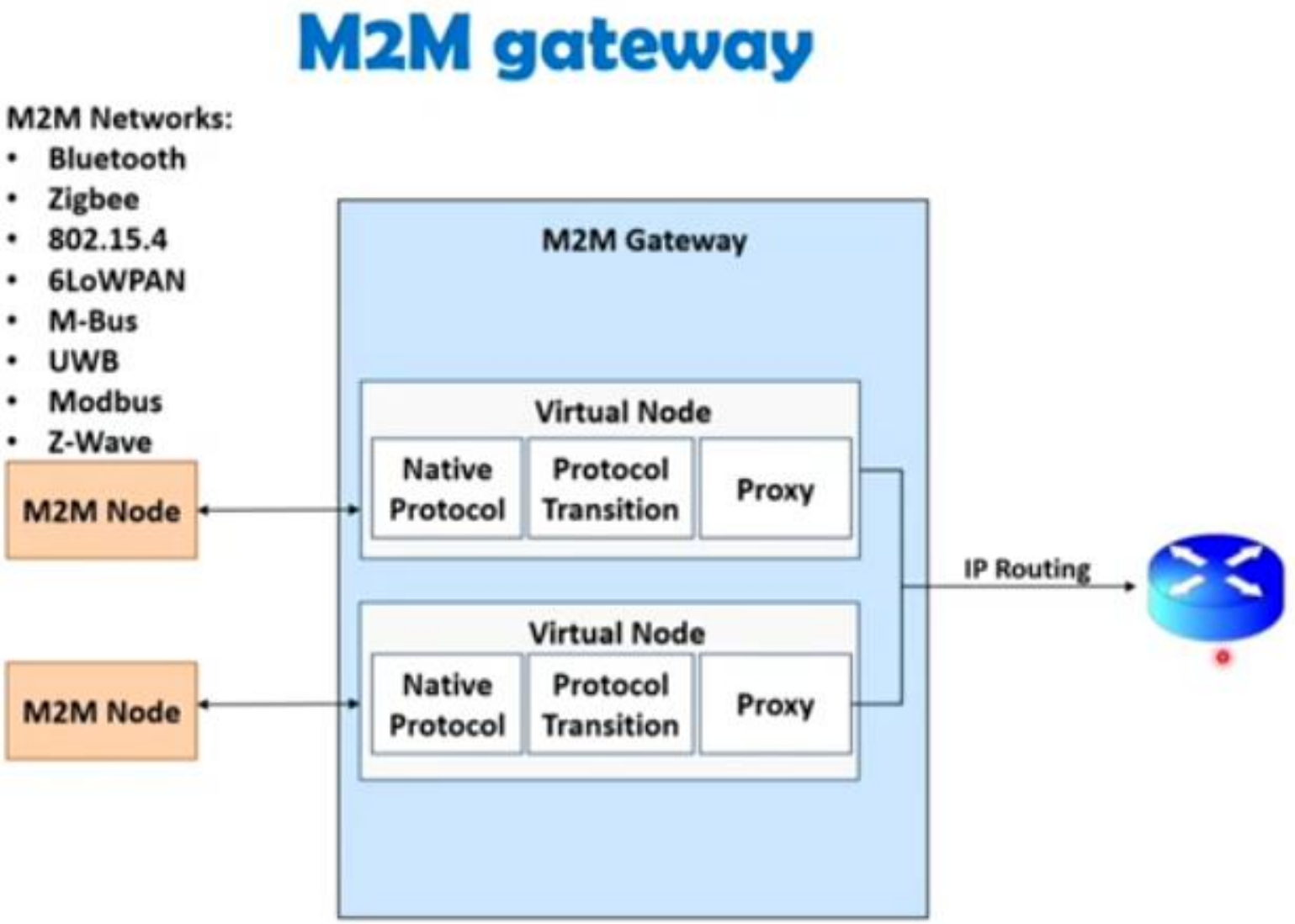


M2M & IoT-A basic Perspective-Introduction

- **What is M2M gateway?**
- M2M gateway acts as an intermediatory between various devices and systems in a machine-to-machine setup.
- It provides the following functionalities:
  - Connectivity between machines.
  - Data exchange between machines.
  - Compatibilities between different networks.

- **What is the purpose or necessity of an M2M gateway?**
- Within the network nodes communicate with each other.
- To communicate with remote M2M area network, M2M gateway is required.

M2M & IoT-A basic Perspective-Introduction



M2M & IoT-A basic Perspective-Introduction

Key Application areas of M2M

TELEMATICS

Connected cars used for safety and security, services and infotainment.



METERING

Meters to report consumption, mainly electricity.



REMOTE MONITORING

Sensors connected to assets are tracked and monitored in real-time.



FLEET MANAGEMENT

Vehicles can be managed and tracked through the path they go.



SECURITY

Connectivity used for home and small business security alarms.

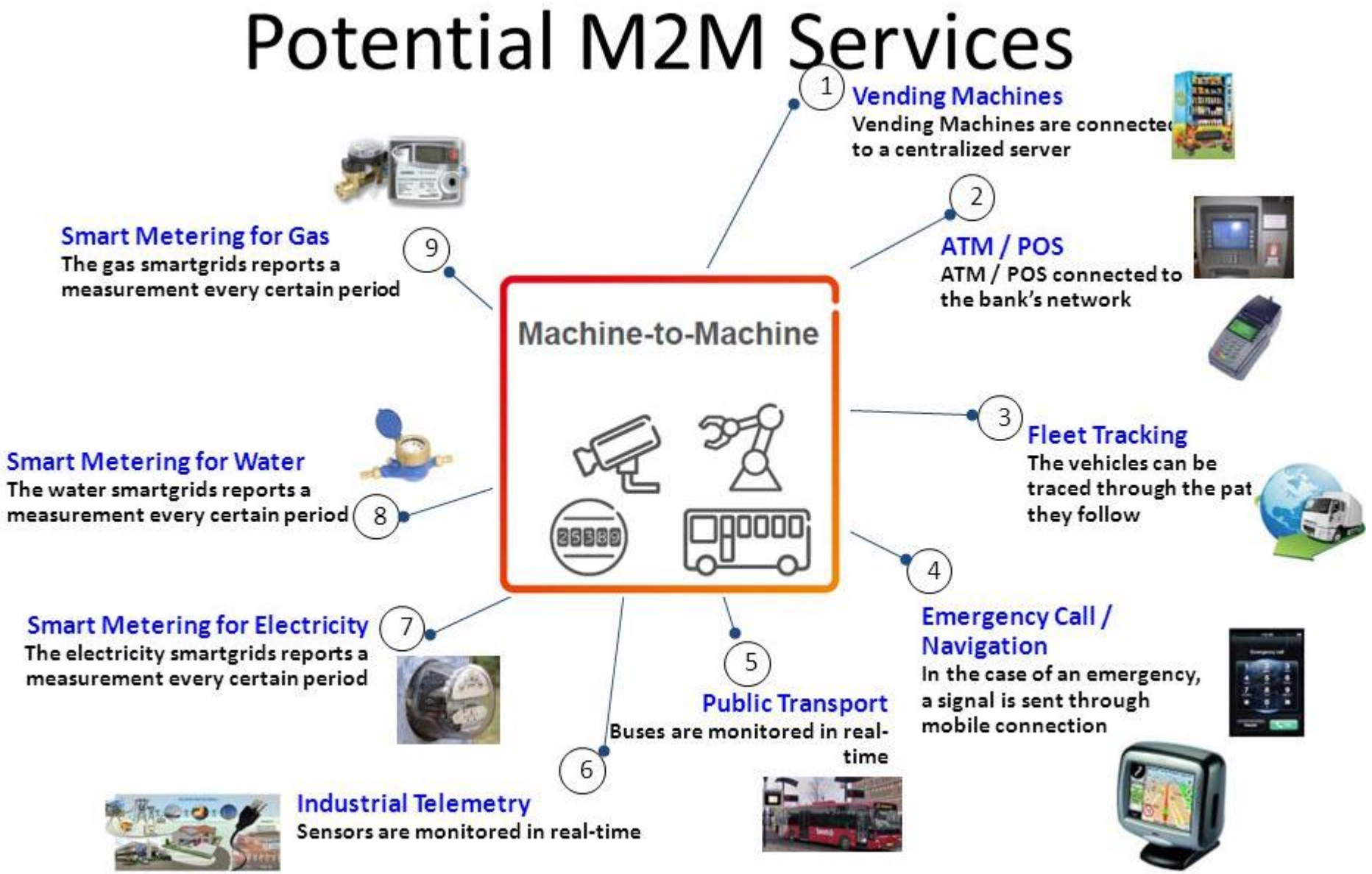


ATM / POINT OF SALES

ATM and POS devices are connected to a centralized secure environment.



M2M & IoT-A basic Perspective-Introduction



M2M & IoT-A basic Perspective-Introduction

What is IoT ?

IoT is a **network of physical objects** — embedded with sensors, software, and connectivity — that collect and exchange data over the internet, often reaching cloud platforms and human-facing applications.

Building Blocks of an IoT System

Every IoT deployment — however complex — is built from these four elements.



1. Sensing / Actuation

Sensors capture temperature, motion, light, etc.; actuators act on the physical world (motors, valves, switches).



2. Connectivity

Wi-Fi, Bluetooth, Zigbee, LPWAN, or cellular links carry data from devices to gateways or the cloud.



3. Data Processing

Edge devices or cloud platforms clean, store, and analyse the incoming data streams.



4. User Interface

Dashboards, mobile apps, or automated triggers turn insight into visible, actionable output.

Differences and similarities between M2M & IoT

Similarities

- Both involve **machines/devices communicating with each other** without constant human intervention.
- Both use **sensors and communication technology** to collect and transmit data.
- Both aim to **automate processes** and enable data-driven decision-making.
- Both can use similar underlying communication protocols (cellular, Wi-Fi, Zigbee, etc.)
- IoT is often considered an **evolution/superset of M2M** — they share the same foundational concept of machines exchanging data.

Differences and similarities between M2M & IoT

Differences

Aspect	M2M	IoT
Definition	Direct communication between two machines/devices, typically point-to-point	Network of interconnected devices communicating over the internet, often through cloud platforms
Communication type	Point-to-point (device to device), often via proprietary or hardwired links	Device to cloud, device to device, and device to gateway — networked and internet-based
Internet dependency	Not necessarily internet-based — can use direct links, cellular, or private networks	Relies heavily on internet connectivity (IP-based networking)

Differences and similarities between M2M & IoT

Differences

Aspect	M2M	IoT
Scale	Usually limited/isolated systems (e.g., a vending machine reporting stock to a central server)	Massive scale — millions of interconnected devices globally
Data sharing	Data is usually used by a single application or a closed system	Data can be shared and accessed by multiple applications/stakeholders via the cloud
Protocols used	SNMP, proprietary protocols, cellular (GSM/GPRS)	HTTP, MQTT, CoAP, WebSocket, and other internet-standard protocols

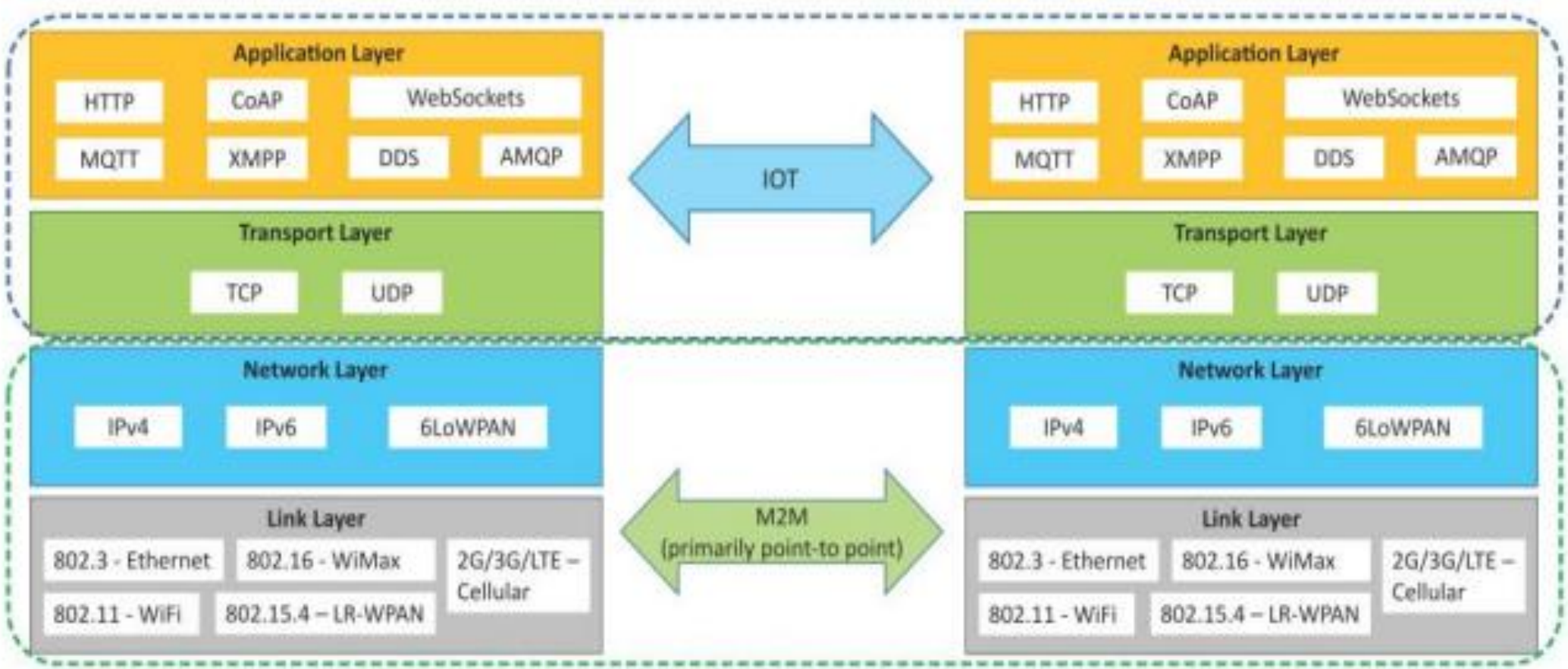
Differences and similarities between M2M & IoT

Differences

Aspect	M2M	IoT
Analytics/intelligence	Limited — mainly data transfer, minimal processing at the machine level	Strong emphasis on cloud-based analytics, AI/ML, big data processing
Human interaction	Minimal to none — machines act independently	Often includes human interfaces (apps, dashboards) for monitoring/control
Examples	ATM reporting cash levels to bank server, vending machine inventory reporting	Smart home systems, wearable fitness trackers, smart city infrastructure, connected cars
Interoperability	Limited — devices often communicate within a specific closed system	High — designed to work across various platforms and device types

Differences and similarities between M2M & IoT

Communication in IoT vs M2M



SDN for IoT

SDN- Software Defined Networking

THE PROBLEM

Why traditional networks strain under IoT

1

Scale

Billions of heterogeneous devices join and leave networks constantly.

2

Heterogeneity

Diverse protocols (Zigbee, BLE, MQTT, CoAP) and hardware capabilities.

3

Dynamic topology

Mobile and battery-constrained nodes change position and availability.

4

Static configuration

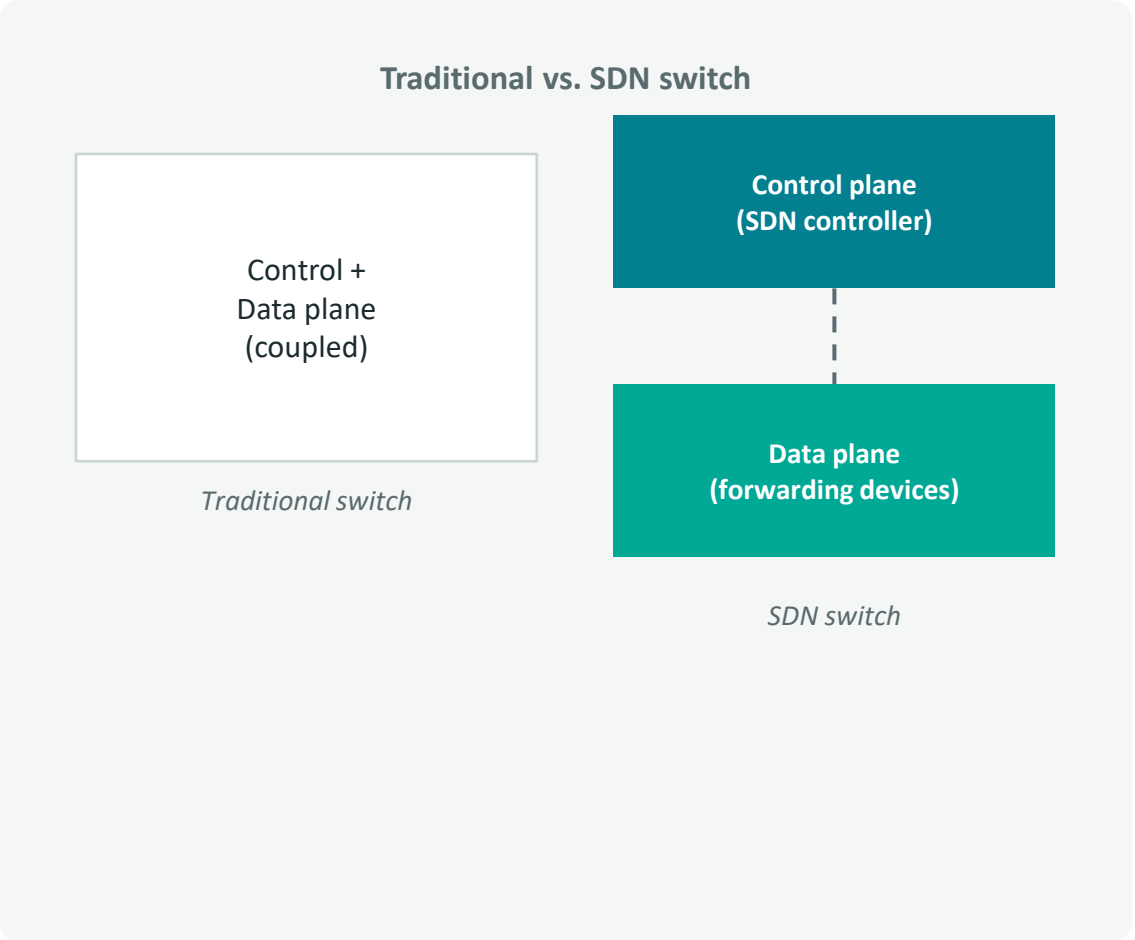
Traditional switches/routers are manually and rigidly configured.

SDN

What is Software-Defined Networking?

SDN separates the **control plane** (decision-making) from the **data plane** (packet forwarding), centralizing network intelligence in software.

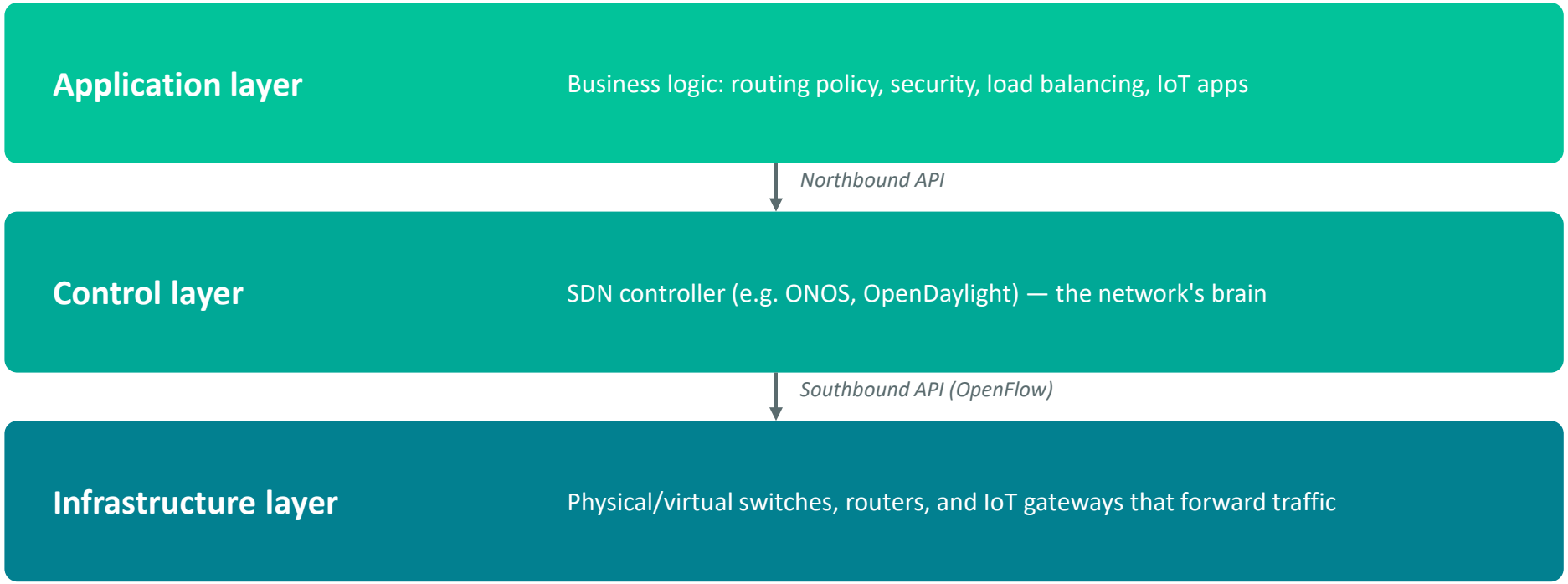
- Devices simply forward packets as instructed
- A central controller sees the whole network
- Behavior is programmable via open APIs (e.g. OpenFlow)
- Enables dynamic, automated network management



SDN

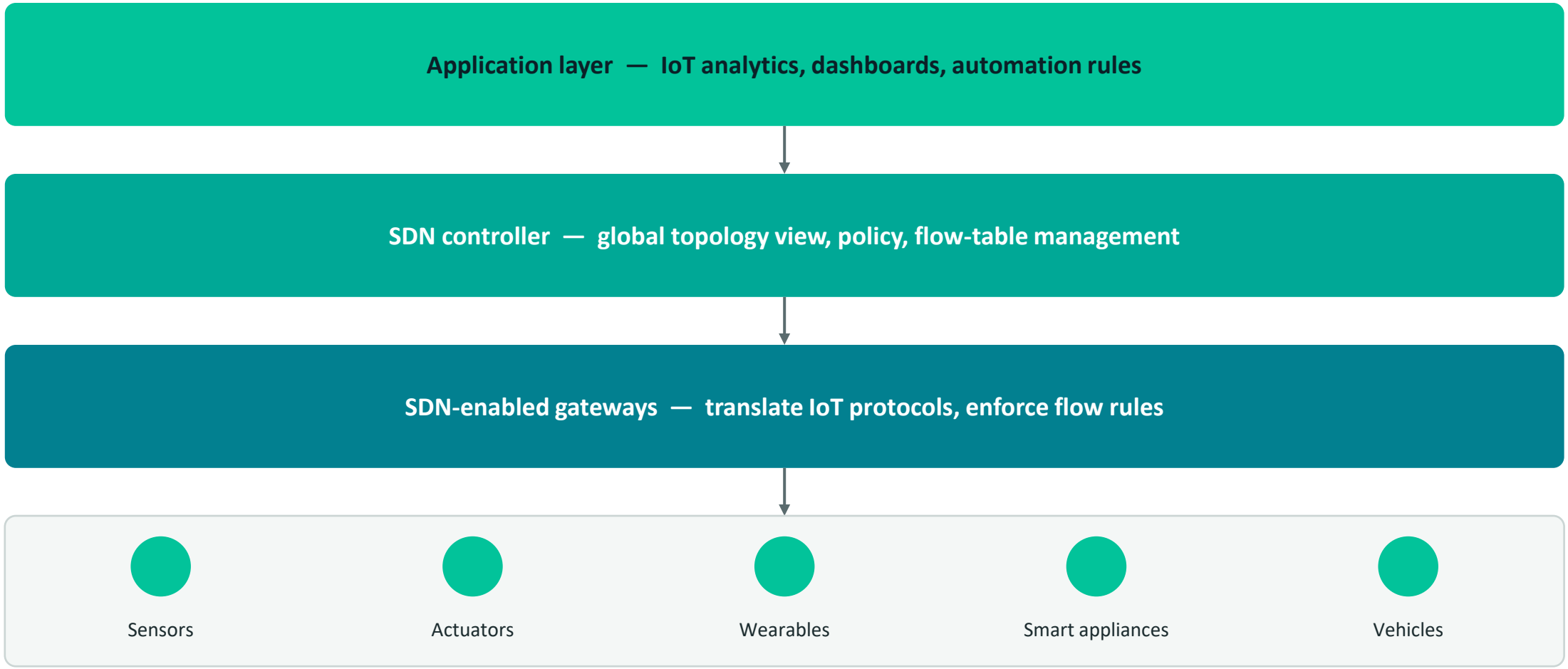
ARCHITECTURE

The three-layer SDN architecture



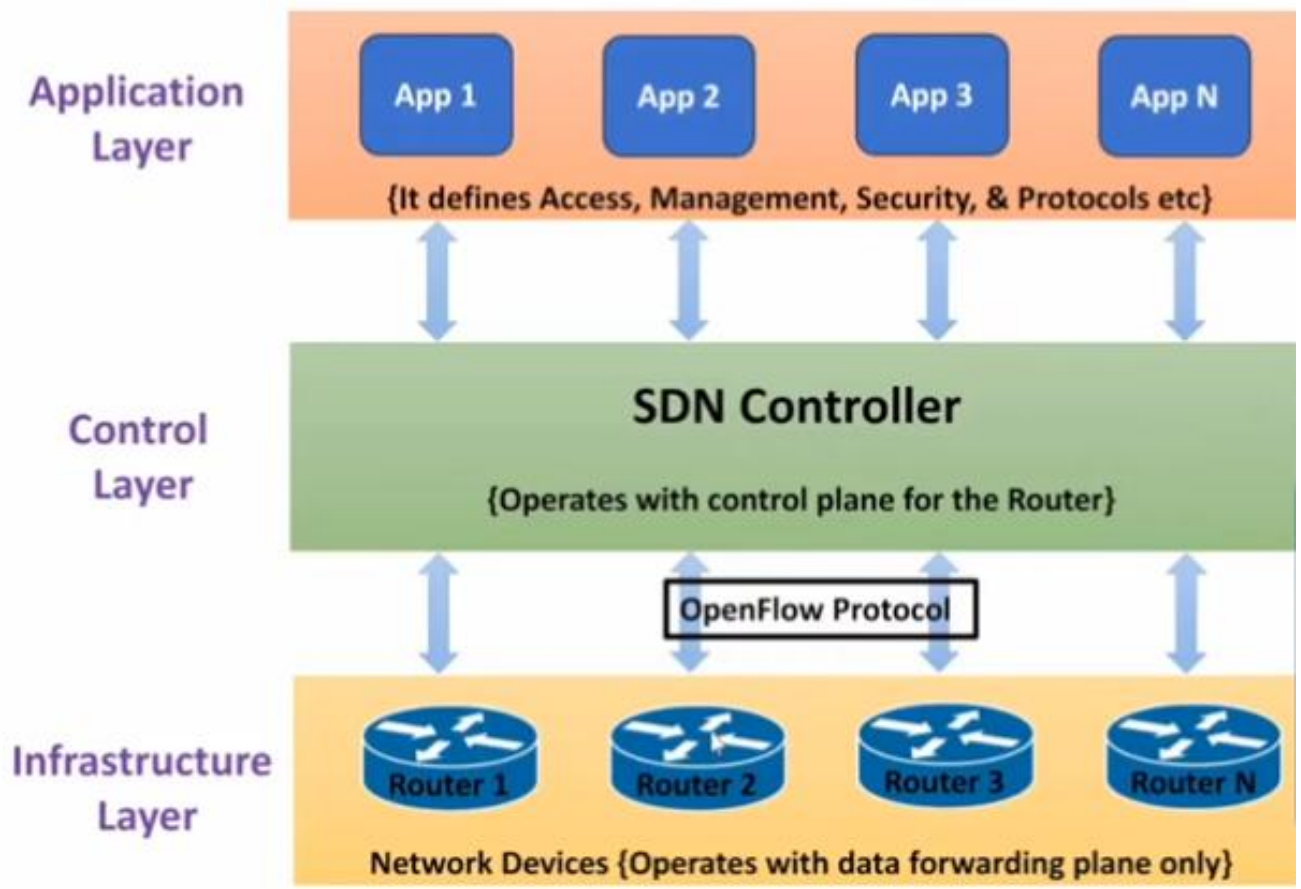
SDN for IoT

A typical SDN-IoT architecture



SDN for IoT

SDN Architecture



SDN for IoT

Why bring SDN into IoT networks?

Centralized visibility

One controller sees every device, flow, and link — simplifying management at IoT scale.

Programmability

Network behavior updates via software, not manual per-device configuration.

Dynamic flow control

Routes and QoS adapt in real time as devices join, move, or fail.

Network slicing

Virtual, isolated networks for different IoT applications on shared infrastructure.

BENEFITS

What SDN brings to IoT deployments

Scalability

Add thousands of devices without reconfiguring each switch manually.

Energy efficiency

Controller can reroute or sleep idle paths to save power on constrained nodes.

Security

Centralized policy enforcement and rapid isolation of compromised devices.

QoS management

Prioritize latency-sensitive traffic (e.g. health monitors) over bulk data.

Interoperability

Controller abstracts diverse IoT protocols into a common control interface.

Rapid innovation

New network services deployed as software, without new hardware.

OPEN ISSUES

Challenges of SDN-based IoT

Single point of failure	A centralized controller is a critical target — needs redundancy/clustering.
Controller overhead	Massive IoT device counts can overwhelm controller processing and flow tables.
Latency	Control-plane round trips may be too slow for real-time IoT control loops.
Security of the controller	Compromising the controller compromises the entire network.
Resource-constrained devices	Many IoT nodes can't run full SDN agents (limited memory/power).

SDN for IoT

APPLICATIONS

Real-world SDN-IoT use cases

Smart homes

Prioritize security-camera streams; isolate a compromised smart-plug automatically.

Smart cities

Dynamically reroute traffic-sensor data during congestion or emergencies.

Industrial IoT

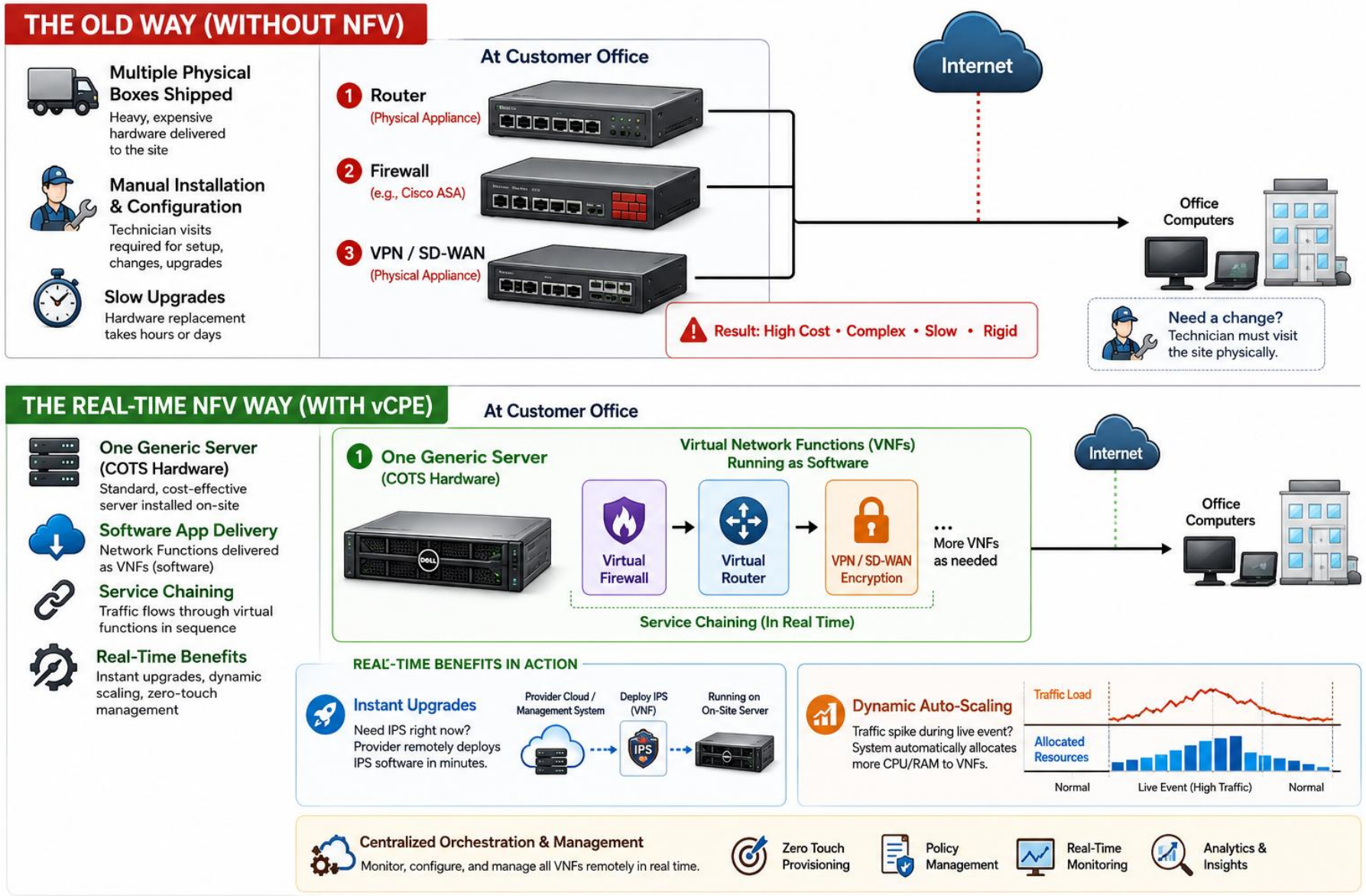
Guarantee low latency for factory-floor control loops via flow prioritization.

Smart healthcare

Isolate and prioritize critical patient-monitoring traffic on shared hospital networks.

NFV for IoT

What Is Network Function Virtualization?



NFV for IoT

What Is Network Function Virtualization?

NFV replaces dedicated network hardware (firewalls, routers, load balancers) with software running on standard servers.

Traditional Approach

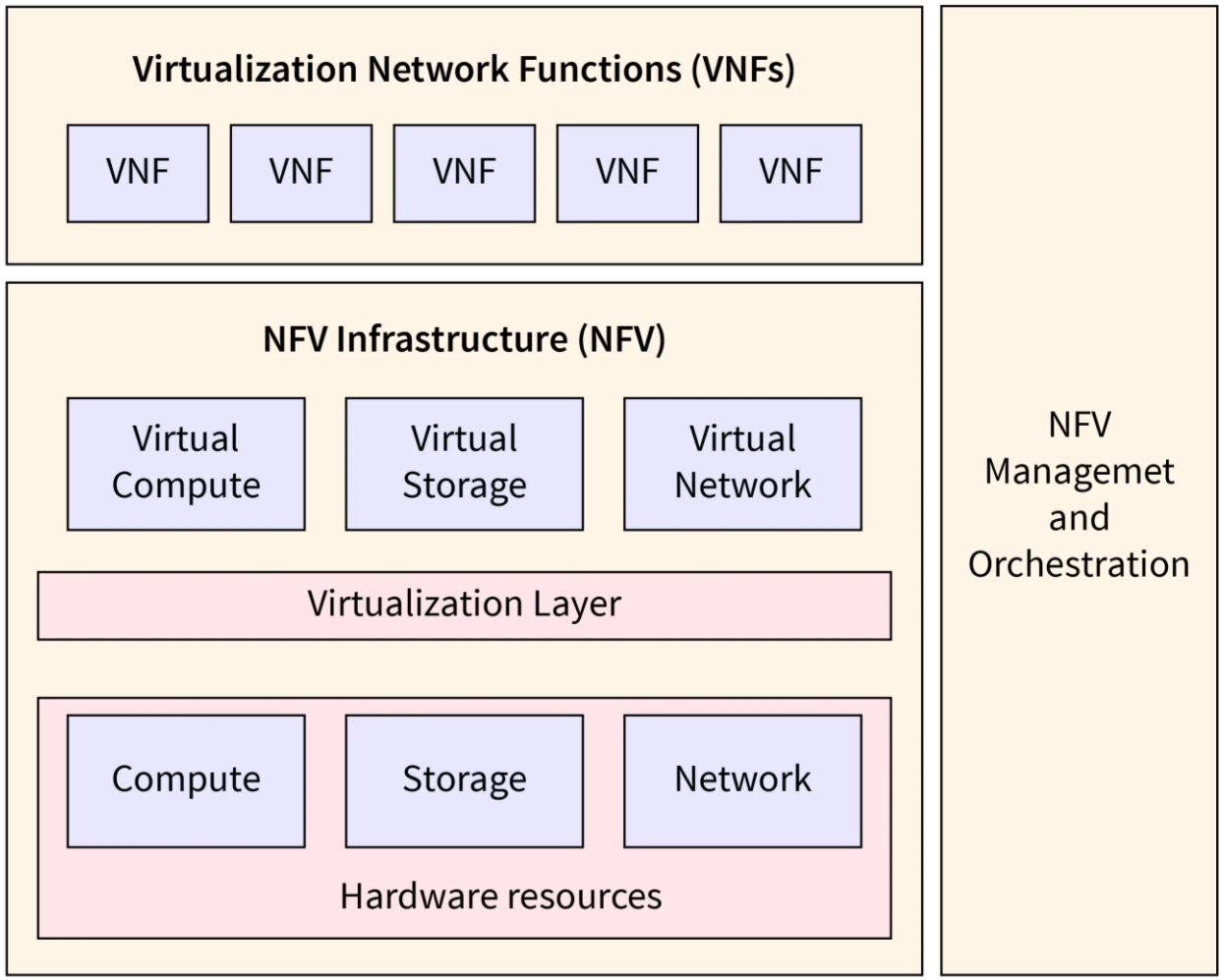
- Purpose-built hardware boxes
- One function per physical device
- Slow to deploy or scale
- High capital expenditure

NFV Approach

- Software functions on shared servers
- Many virtual functions per device
- Rapid deployment via automation
- Lower cost, elastic scaling

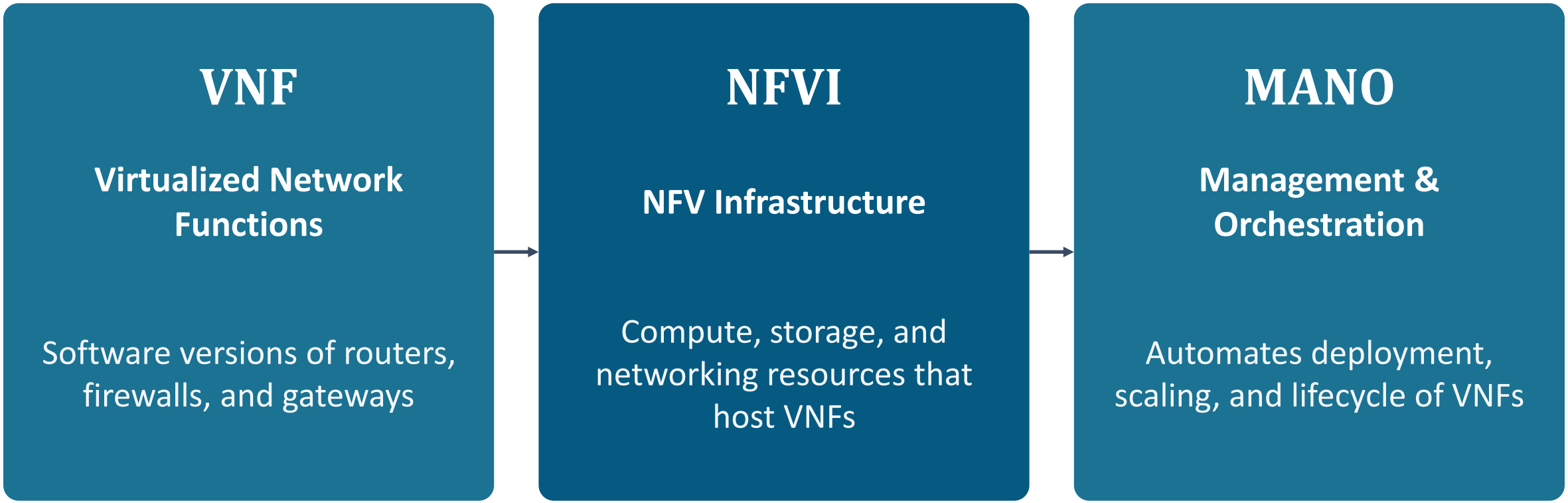
NFV for IoT

NFV Architecture



NFV for IoT

Core NFV Architecture



NFV for IoT

Key Elements of NFV

- **Virtualized Network Functions (VNFs):** Software-based network appliances (e.g., virtual firewalls, routers, load balancers) running inside virtual machines or containers instead of dedicated hardware.
- **NFV Infrastructure (NFVI):** The underlying environment that abstracts physical servers, storage, and networking into dynamic virtual resources using a virtualization layer (hypervisor).
- **NFV Management and Orchestration (MANO):** The vertical control plane that provisions, manages, scales, and orchestrates the lifecycle of VNFs and virtual resources.

NFV for IoT

Why Bring NFV to IoT?

IoT deployments scale to millions of devices — NFV provides the elasticity and automation that static hardware cannot.

Scalability

Spin up or down virtual functions as device counts change

Cost Efficiency

Commodity hardware replaces costly proprietary appliances

Flexibility

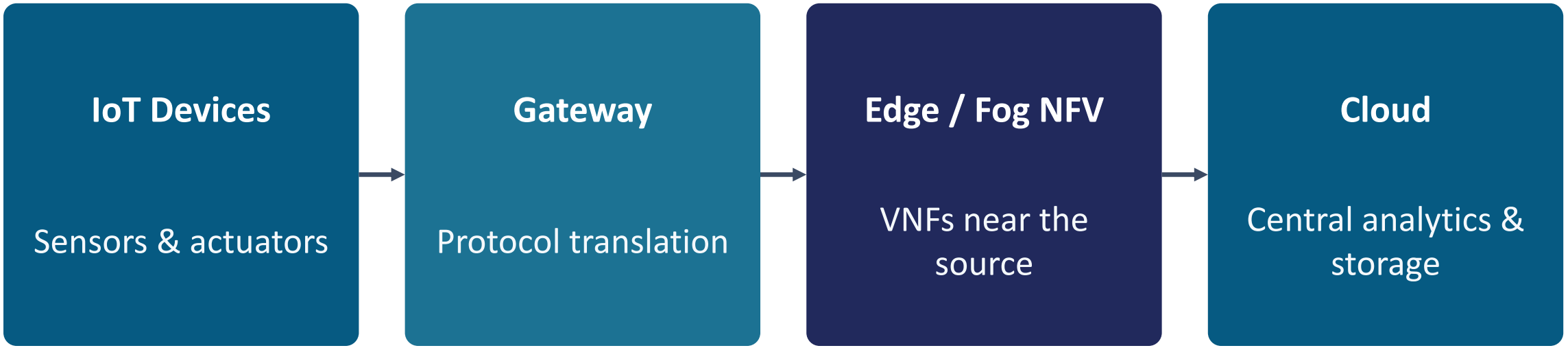
New services deployed as software, not new equipment

Faster Rollout

Automated orchestration cuts deployment time drastically

NFV for IoT

An NFV-Enabled IoT Architecture



Placing VNFs at the edge/fog layer reduces latency for time-critical IoT applications while the cloud retains heavy analytics.

NFV for IoT

Key Enabling Technologies

1. **Software-Defined Networking — SDN**

   Separates control and data planes for programmable traffic management

2. **Hypervisors & Containers — Virtualization**

   Runs multiple isolated VNF instances on shared hardware

3. **Compute Near the Source — Edge / Fog Computing**

   Brings processing closer to IoT devices to cut latency

NFV for IoT

Real-World Use Cases

Smart City

Virtualized traffic and utility management functions scale with sensor density

Smart Home

Home gateways run security and automation VNFs without extra appliances

Industrial IoT

Factory floors deploy monitoring VNFs that scale with production lines

Connected Health

Wearable data is filtered by edge VNFs before reaching the cloud

NFV for IoT

Open Challenges & Research Issues

- **Latency** — Real-time IoT control loops demand very low processing delay
- **Security** — Virtualized functions widen the attack surface across shared infrastructure
- **Resource Limits** — IoT devices and edge nodes have limited compute and energy
- **Orchestration Complexity** — Managing millions of devices and VNFs is a hard scheduling problem
- **Standardization** — Interoperability across vendors and protocols remains incomplete

NFV for IoT

Future Directions

1. **5G / 6G Integration**

   Network slicing tailored to diverse IoT traffic classes

2. **AI-Driven Orchestration**

   Machine learning predicts demand and auto-scales VNFs

3. **Edge-Native NFV**

   Lightweight VNFs designed specifically for constrained edge nodes

4. **Zero-Trust Security**

   Continuous verification across virtualized IoT functions

Primary Deployment Domains



Data Centers (SDN)

Automates multi-tenant cloud networks, micro-segmentation, and dynamic bandwidth allocation.



Telecom Cores (NFV)

Replaces physical mobile packet cores, IMS, and RAN components in modern 5G networks.



Edge & Security (Both)

Deploys virtual firewalls and SD-WAN CPE appliances at enterprise edge branch sites.

SDN vs. NFV: Executive Comparison

Software-Defined Networking reorganizes network control, while Network Functions Virtualization modernizes service appliance hardware.

NETWORK ARCHITECTURE

Software-Defined Networking (SDN)

- **Primary Focus:** Decouples control plane (routing logic) from the data plane (packet forwarding).
- **Key Abstraction:** Centralizes network decision-making into a programmable software controller.
- **Target Environment:** Data centers, enterprise LAN/WANs, and cloud traffic routing.
- **Core Protocols:** OpenFlow, NETCONF, RESTful Southbound & Northbound APIs.
- **Primary Benefit:** Dynamic global traffic management, network-wide automation, and agility.

APPLIANCE VIRTUALIZATION

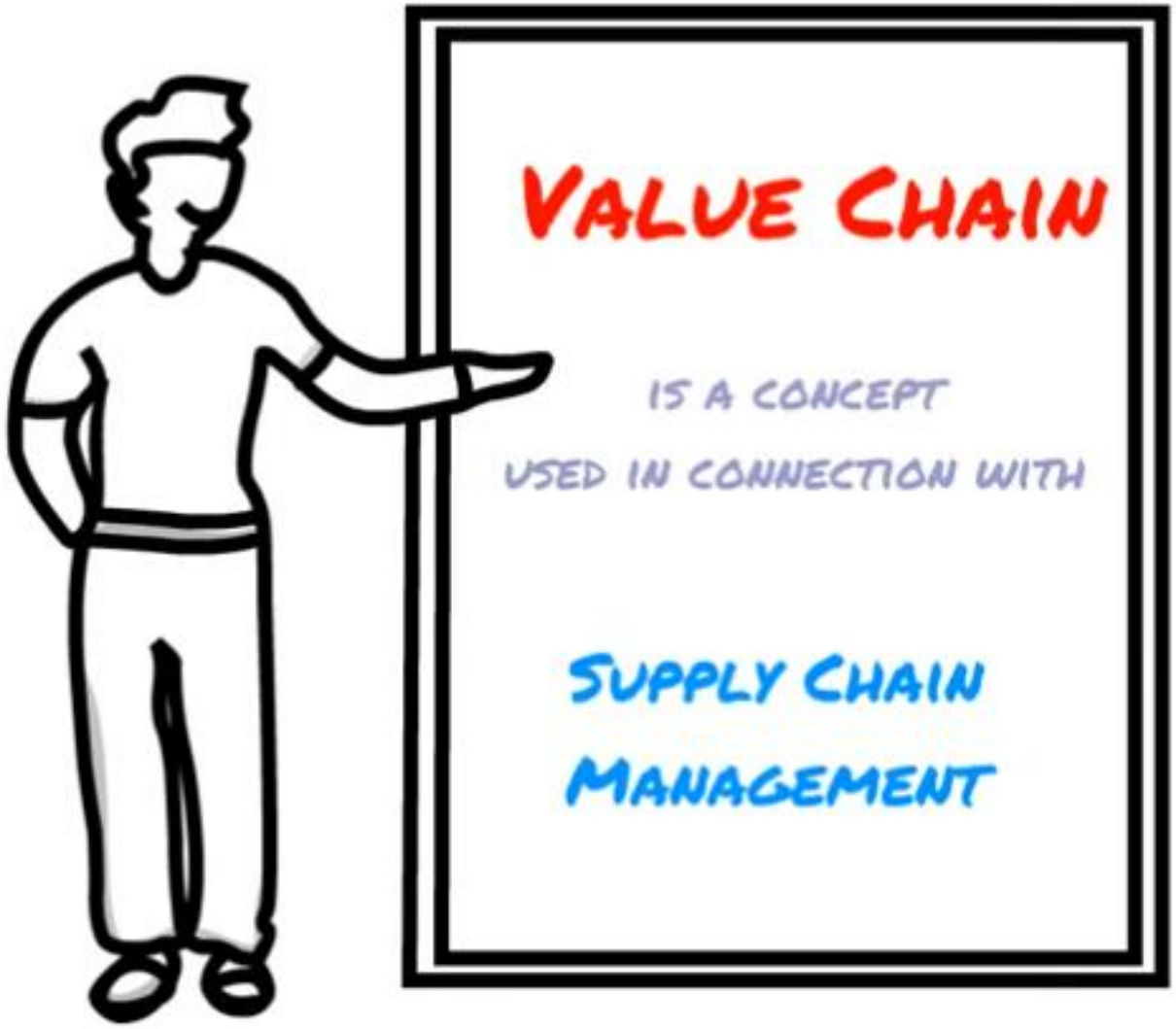
Network Functions Virtualization (NFV)

- **Primary Focus:** Replaces dedicated proprietary hardware appliances with virtual software functions (VNFs).
- **Key Abstraction:** Decouples network software services from proprietary underlying hardware.
- **Target Environment:** Telecom operator cores, edge networks, and service provider infrastructures.
- **Core Standards:** ETSI NFV ISG Framework, MANO Orchestration, Hypervisors/KVM.
- **Primary Benefit:** Eliminates vendor hardware lock-in, reduces CapEx/OpEx, and speeds up deployments.

Synergistic Relationship in Modern Cloud & Telecom

SDN and NFV are highly complementary: **NFV** virtualizes core network services (e.g., Firewalls, Routers, Load Balancers), while **SDN** provides the dynamic, programmable routing required to connect traffic seamlessly between those virtual functions.

Value Chain



It considers
the complete value
of a product
in an
extended supply chain

Value Chain


VALUE CHAIN CONSIDERS
ALL ACTIVITIES INVOLVED IN
CREATING A PRODUCT
OR
PROVIDING A SERVICE.

Value Chain

For a goods manufacturer

Value Chain consists of



Research & Development

Product Design



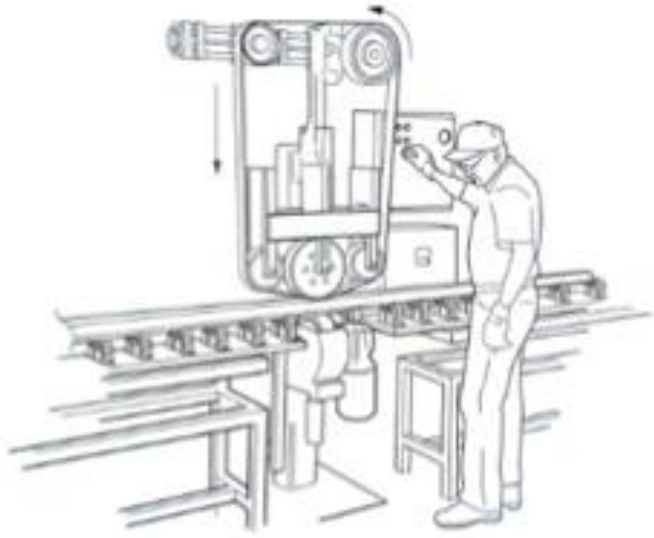
Value Chain

Goods manufacturer's Value Chain



Raw materials purchase

Production



Value Chain

Goods manufacturer's Value Chain



Sales and Marketing

Delivery



Value Chain

Goods manufacturer's Value Chain





and



Service

Customer Relations

Value Chain

Value chain Recap

R&D
Product Design



Raw materials purchase



Production



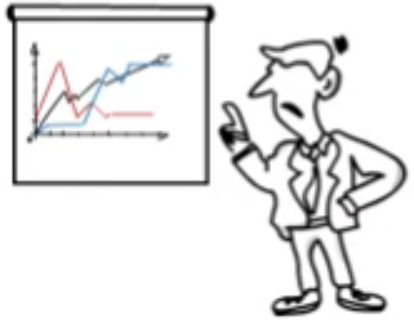
Service
Customer Relations



Delivery



Sales & Marketing



Value Chain

Difference from Supply Chain

A Value Chain is broader in scope than a Supply Chain

A Supply Chain is a portion or a subset of the Vaue Chain that focusses on the phisical movement of material and goods and the related flows of information and financial transactions

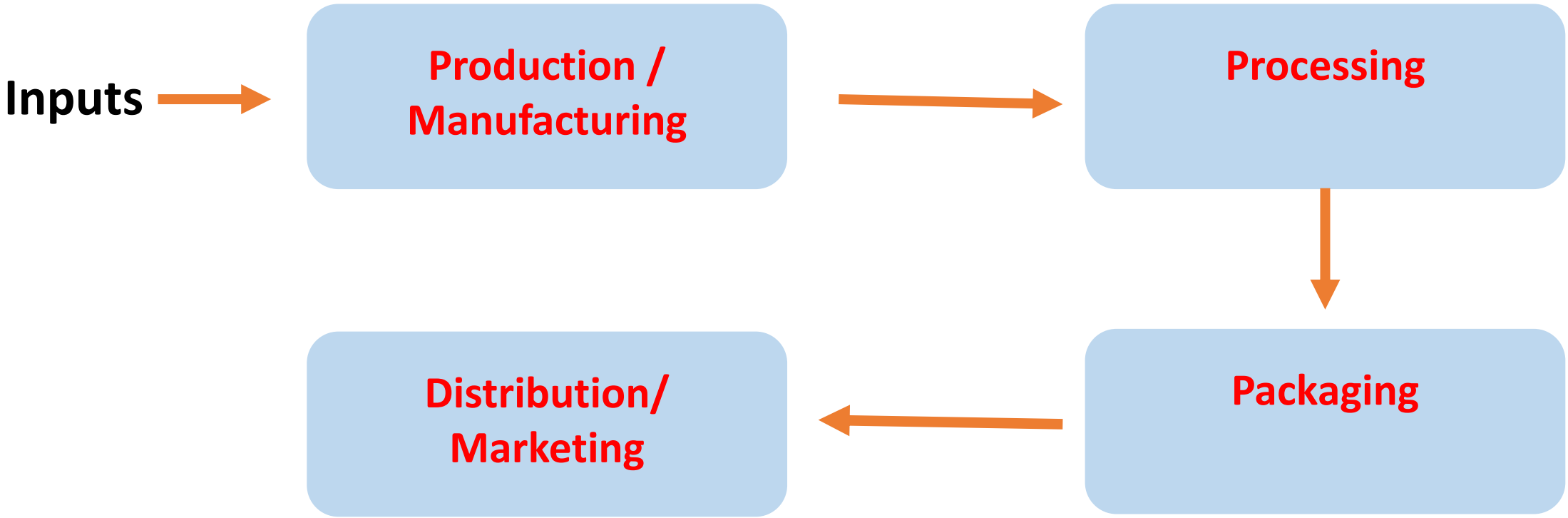
Value Chain

- A value chain describes the full range of activities that firms and workers perform to bring a product from its conception to end use and beyond, including design, production, marketing, distribution, and support to the final consumer.

Value Chain

- Analyzing an industry from a global value chain (GVC) perspective permits understanding of the context of globalization on the activities contained within them by “focusing on the sequences of tangible and intangible value-adding activities, from conception and production to end use.
- GVC analysis therefore provides a holistic view of global industries both from the top down and from the bottom up”.

M2M value Chains



M2M value Chains

M2M value chain Example: Cocoa Beans to Chocolate



M2M value Chains

Input	Inputs are the base raw ingredients that are turned into a product.
Example	cocoa beans for the manufacture of chocolate
M2M example	data from an M2M device that will be turned into a piece of information.

M2M value Chains

Production/ Manufacture:	Production/Manufacture refers to the process that the raw inputs are put through to become part of a value chain.
Example	cocoa beans may be dried and separated before being transported to overseas markets.
M2M example	Data from an M2M, needs to be verified and tagged for provenance.

M2M value Chains

Processing:	Processing refers to the process whereby a product is prepared for sale.
Example	cocoa beans may now be made into cocoa powder, ready for use in chocolate bars.
M2M example	M2M refers to the aggregation of multiple data sources to create an information component something that is ready to be combined with other data sets to make it useful for corporate decision-making.

M2M value Chains

Packaging:	Packaging refers to the process whereby a product can be branded as would be recognizable to end-user consumers.
Example	A chocolate bar would now be ready to eat and have a red wrapper with the words "KitKatt" on it.
M2M example	M2M data will have to be combined with other information from internal corporate databases, for example, to see whether the data received requires any action. This data would be recognizable to the end-users that need to use the information, either in the form of visualizations or an Excel spreadsheet.

M2M value Chains

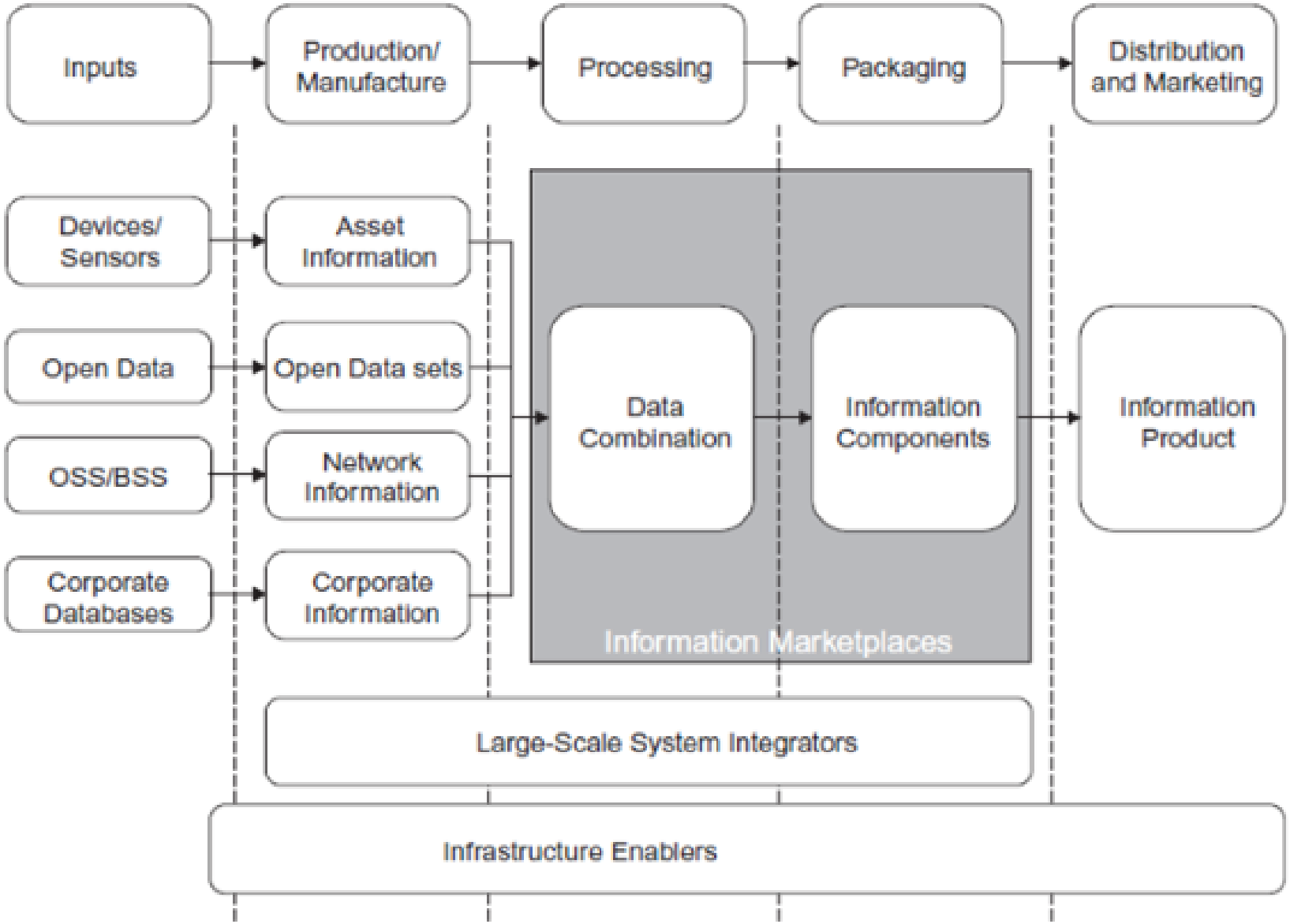
Distribution/ Marketing:	This process refers to the channels to market for products.
Example	a chocolate bar may be sold at a supermarket, a kiosk, or even online.
M2M example	will have produced an Information Product that can be used to create new knowledge within a corporate environment examples include more detailed scheduling of maintenance based on real-world information or improved product design due to feedback from the M2M solution.

Unit-2- part-I

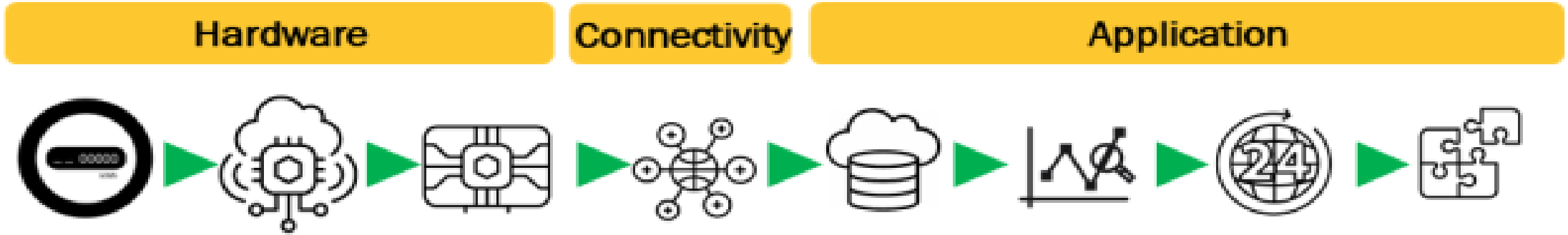
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IoT value Chains



IoT value Chains



Device	Sensors	Comms hardware	Connectivity	Backend	Software	Billing and support	Systems integration
The device or machine (for example, a smart meter)	Sensors and actuators that capture the data and perform the changes to the device	The SIMs, eUICCs and iSIMs, modules and chipsets that connect the device to the network	Cellular (2G, 3G, 4G, NB-IoT, LTE-M) LoRa Wi-Fi Bluetooth Satellite Fibre Wi-SUN	The servers to collect and analyse the data	Application develop-ment Application enable-ment Device manage-ment Data analytics	Billing and charging Customer support	Integration of the key network components with the backend systems.

From M2M to IoT: What Changes

M2M

- Device-to-device links
- Connectivity-centric value
- Closed, proprietary systems
- Linear, telecom-led chain

IoT

- Device-to-cloud-to-user links
- Data & platform-centric value
- Open, interoperable ecosystems
- Networked, layered value chain

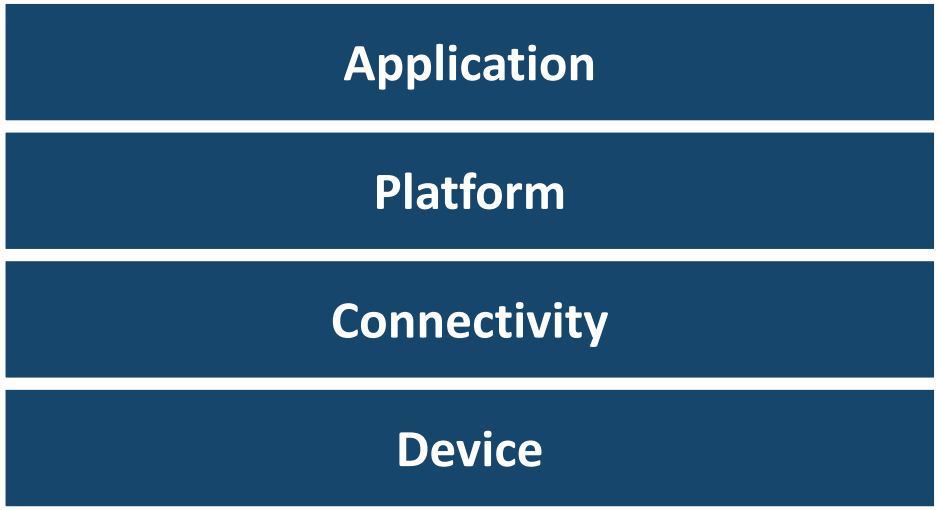
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An Emerging Industrial Structure for IoT

Vertical (Silo) Structure



One vendor owns every layer — closed, proprietary, hard to mix and match.

Horizontal (Layered) Structure



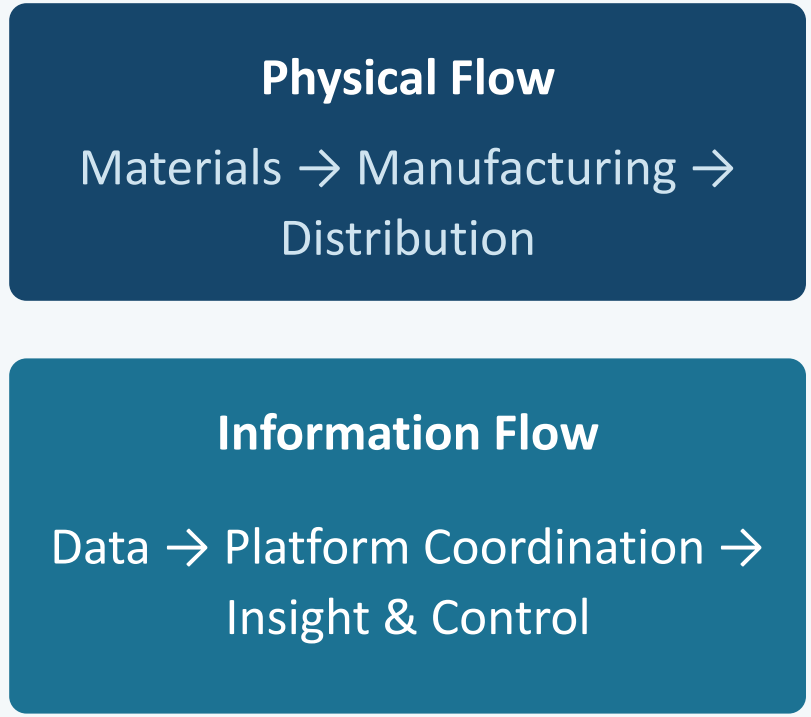
Specialized firms compete within each layer — open, modular, interoperable.

The New Horizontal Layers, Layer by Layer

1	**Device Layer**	Sensor, chipset and hardware manufacturers building the physical endpoints.
2	**Connectivity Layer**	Telecom operators and low-power networks (cellular, Wi-Fi, LPWAN) moving the data.
3	**Platform Layer**	Cloud IoT platforms for device management, storage, and analytics at scale.
4	**Application Layer**	Vertical solution and service providers turning data into business outcomes.

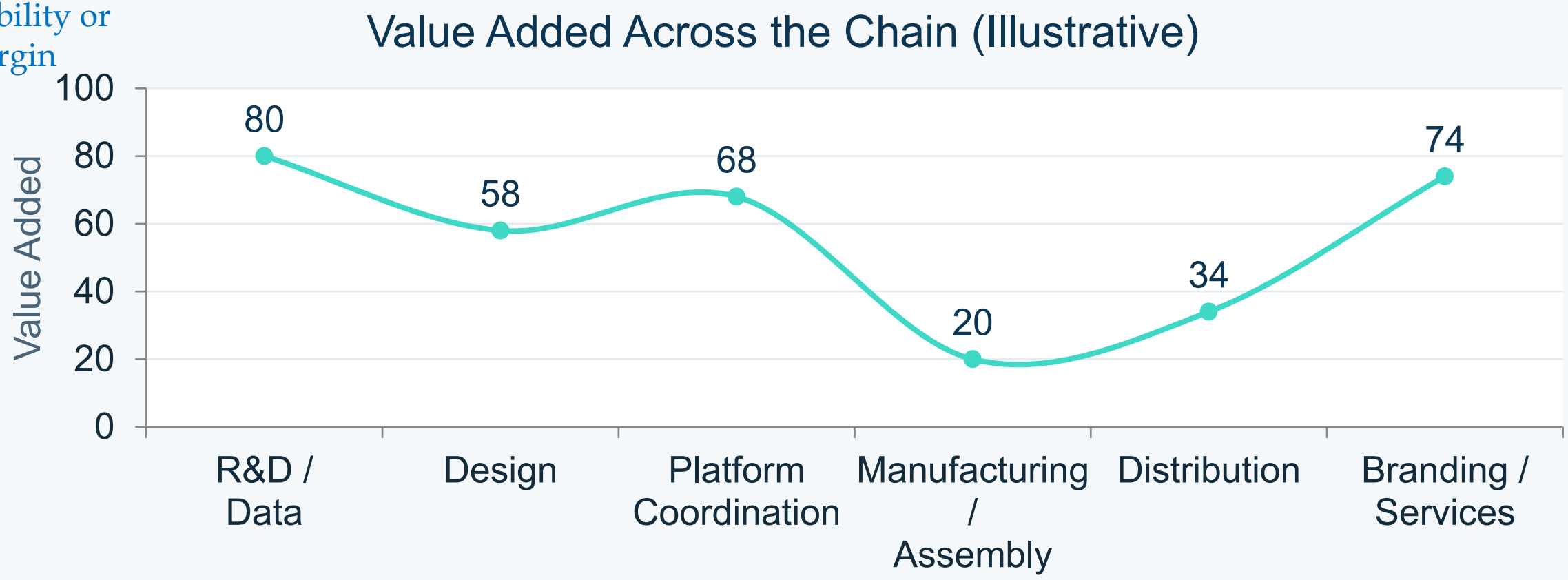
The Information-Driven Global Value Chain

- Traditional GVCs: value comes from globally dispersed production and manufacturing tasks.
- Information-driven GVCs: value increasingly comes from data flows, digital intelligence and platform coordination layered on physical production.
- The “chain” becomes a network coordinated by information — not only by the movement of goods.



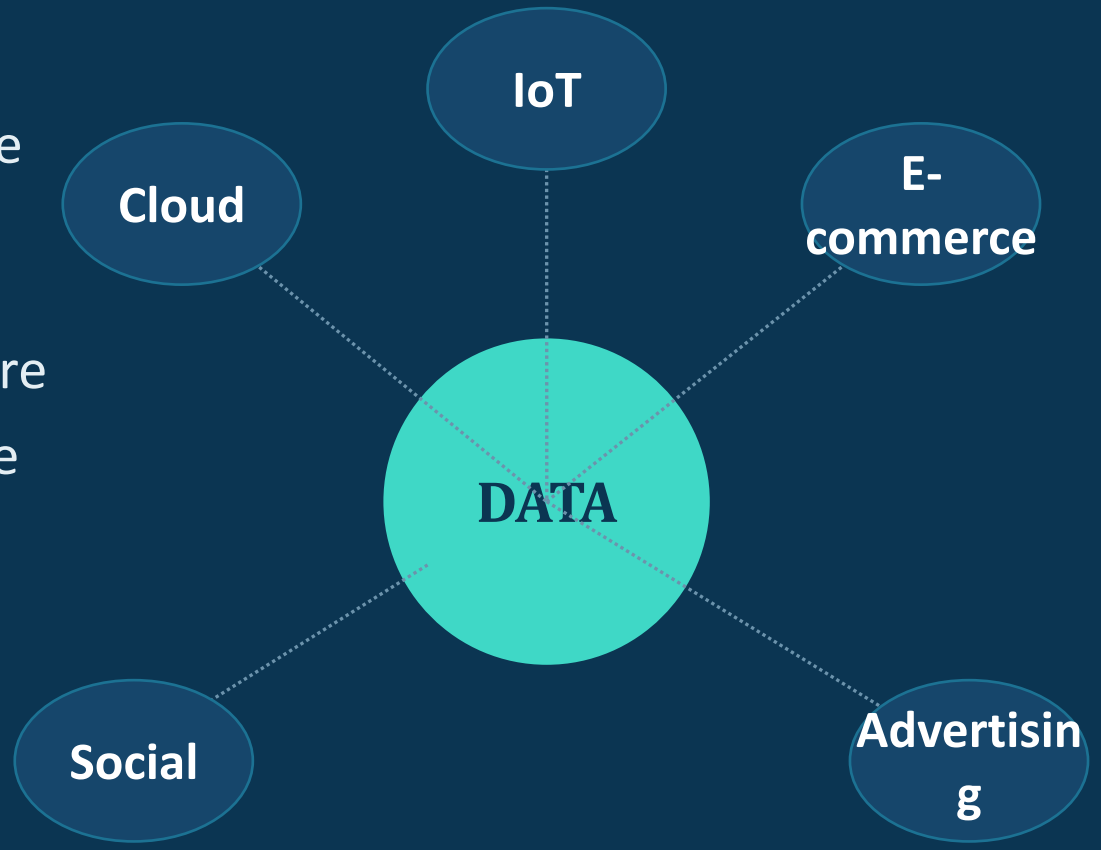
Where the Value Lies: The Smile Curve

Relative profitability or margin



Global Information Monopolies

- A small number of global platform firms concentrate data, infrastructure and digital market power.
- Driven by network effects: more users generate more data, which improves services and attracts still more users.
- Their control spans several layers at once — cloud, platforms, applications and advertising.



Implications & Policy Responses

Implications

- High entry barriers for smaller firms and nations
- Data becomes a strategic economic & geopolitical asset
- Decision-making power concentrates in a few firms

Policy Responses

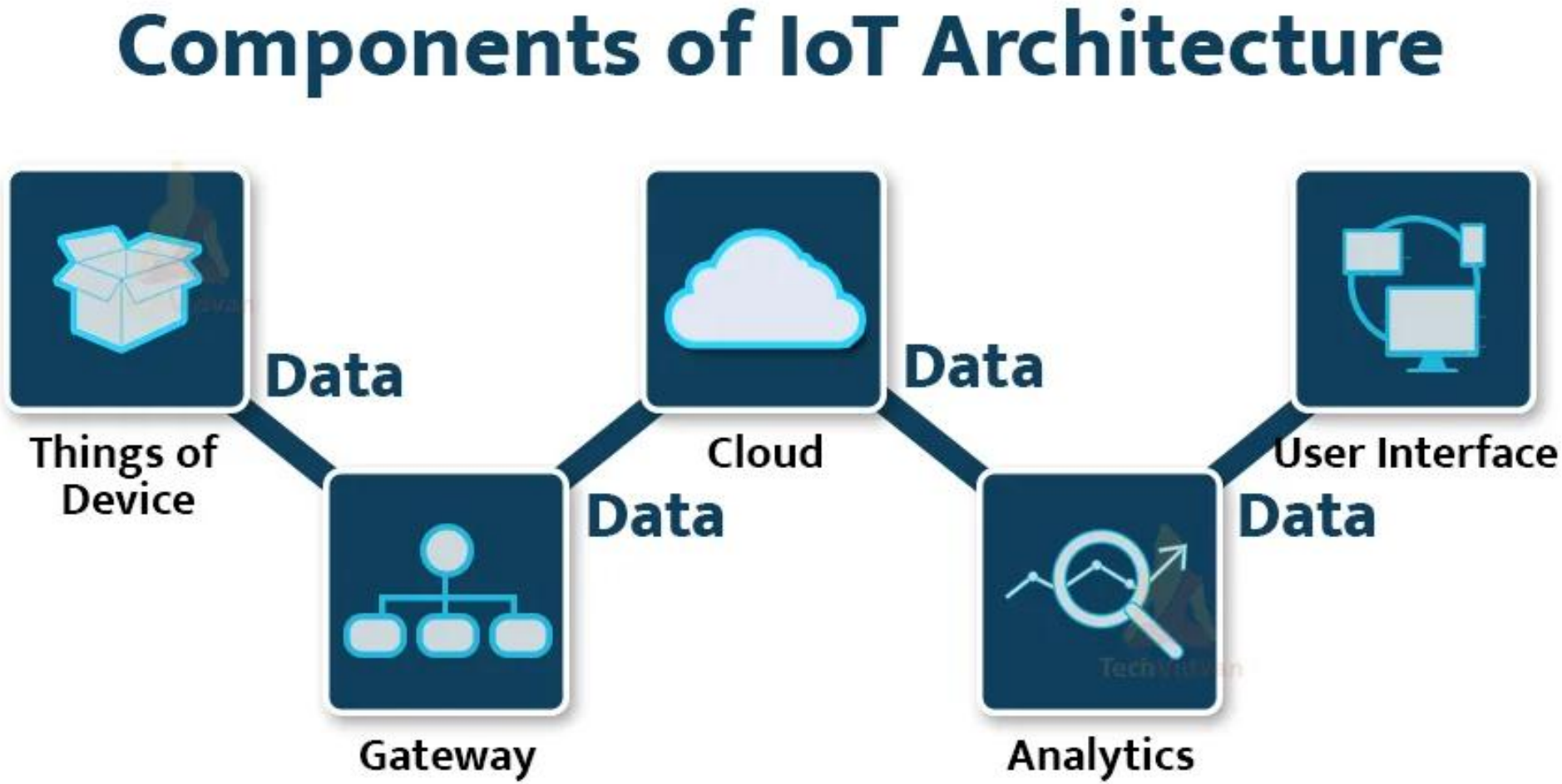
- Data localization requirements
- Competition and antitrust regulation
- Digital sovereignty initiatives

Unit-2- part-II

IoT Architecture

- IoT Architecture Components
- Comparing IoT Architectures
- A Simplified IoT Architecture
  - Core IoT functional Stack
  - IoT data management and compute stack

IoT Architecture Components



IoT Architecture Components

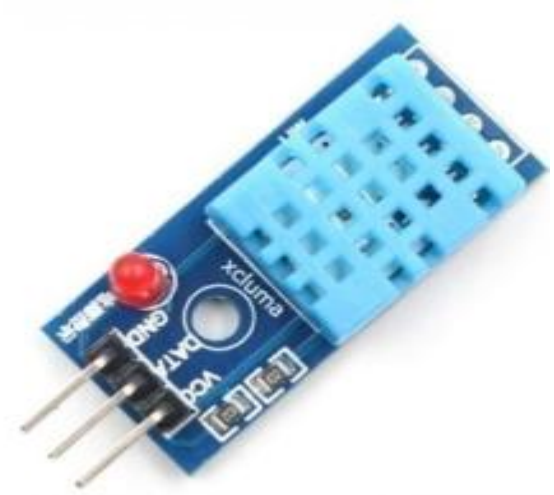
IoT (Internet of Things) architecture typically consists of several key components that work together to enable devices to communicate, process data, and provide insights.

Here are the main components:

1. **Devices/Sensors**
2. **Connectivity**
3. **Edge Computing**
4. **Cloud Infrastructure**
5. **Data Management and Analytics**
6. **User Interface**
7. **Security**
8. **Application Layer**
9. **Protocols**
10. **Management and Orchestration**

IoT Architecture Components

1. **Devices/Sensors**:
   Physical devices or sensors that collect data from the environment (e.g., temperature sensors, cameras, smart appliances).







IoT Architecture Components

2. Connectivity:

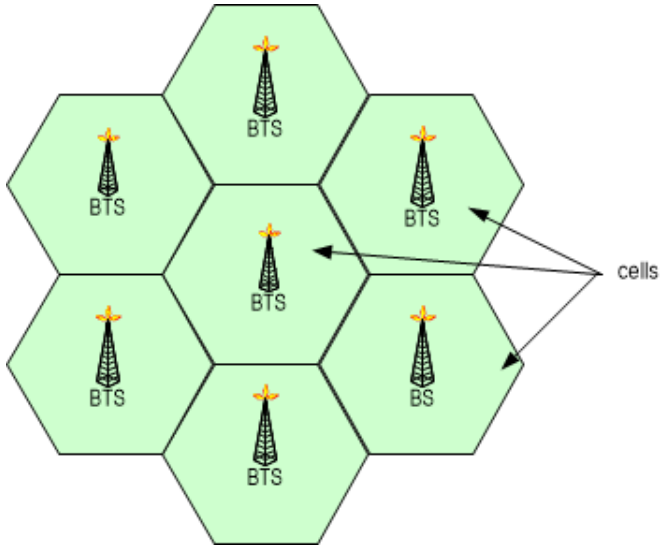
Communication protocols that enable devices to connect to the internet or to each other (e.g., Wi-Fi, Bluetooth, Zigbee, LoRaWAN, cellular networks).









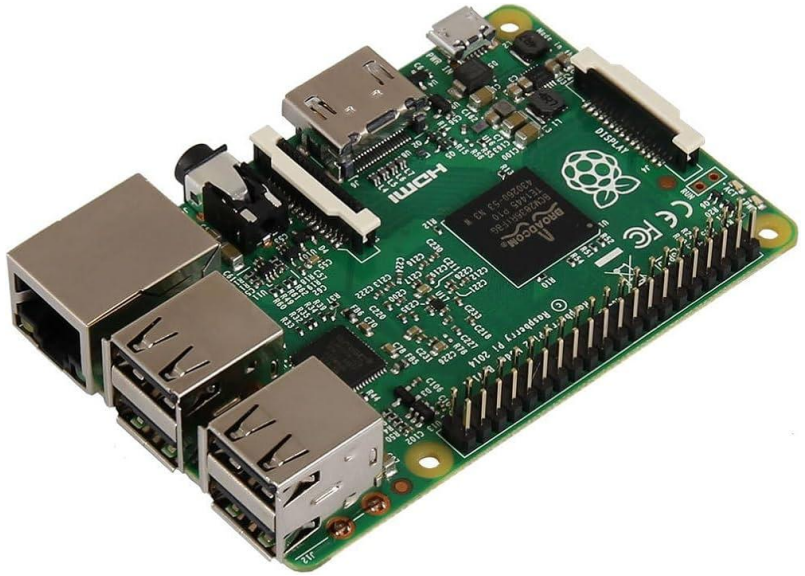


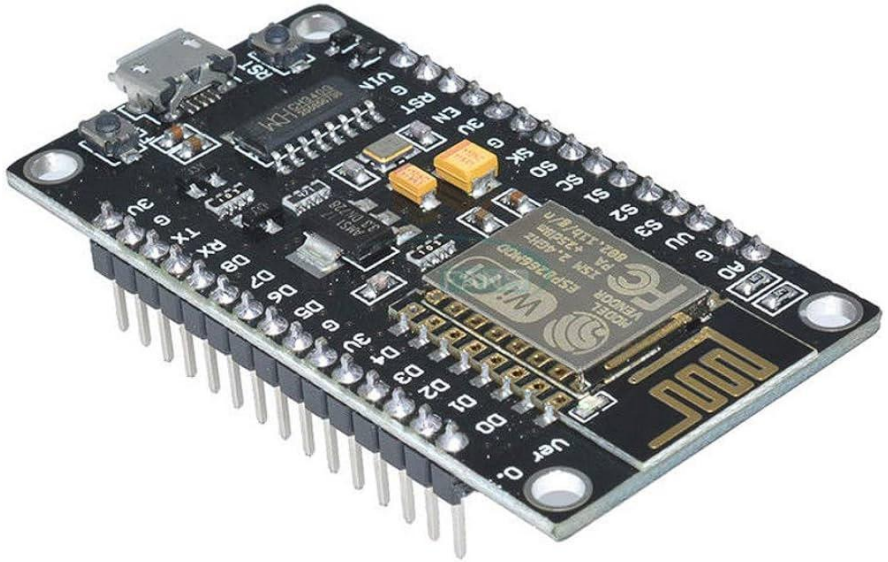
IoT Architecture Components

3. Edge Computing:

Processing data near the source (at the "edge") to reduce latency and bandwidth use.

Edge devices can filter, aggregate, and preprocess data before sending it to the cloud.





IoT Architecture Components

4. Cloud Infrastructure:

- Centralized servers and services that store, process, and analyze data collected from IoT devices.
- This often includes data storage, machine learning capabilities, and application hosting.

ThingSpeak



AWS IoT Core

aws

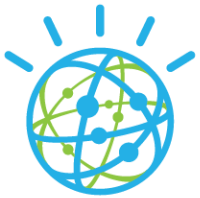
IoT Architecture Components

5. Data Management and Analytics:

- Tools and platforms for managing the data lifecycle, including storage, retrieval, analysis, and visualization.
- This may involve big data technologies, databases, and analytics frameworks.



Google BigQuery



IBM Watson

IoT Architecture Components

6. User Interface:

Front-end applications (web, mobile, dashboards) that allow users to interact with the IoT system, monitor devices, and visualize data.





IoT Architecture Components

7. Security:

Mechanisms to protect data integrity, confidentiality, and availability. This includes encryption, authentication, and access control to secure devices and communications.

Ex:

•TLS (Transport Layer Security) used to encrypt data transmitted between IoT devices and the cloud

•OAuth 2.0 for secure authorization of devices accessing cloud services.



IoT Architecture Components

8. Application Layer:

Software applications that utilize the data generated by IoT devices to provide specific services, such as smart home automation, predictive maintenance, or health monitoring.

Example:

- A smart agriculture application that analyzes soil moisture data to optimize irrigation schedules.
- A predictive maintenance application that analyzes data from industrial machines to forecast failures and schedule maintenance.

IoT Architecture Components

9. Protocols:

Standardized communication protocols (e.g., MQTT, HTTP, CoAP) that facilitate data exchange between devices and servers.

Example: Standardized communication protocols (e.g., MQTT, HTTP, CoAP) that facilitate data exchange between devices and servers.







IoT Architecture Components

10. Management and Orchestration:

Tools for managing devices, deploying applications, monitoring performance, and ensuring compliance with regulations.

Example:

•Azure IoT Hub, which provides tools for device management, monitoring, and data integration for IoT solutions.

•Google Cloud IoT Core, providing device management and data ingestion services for IoT solutions.





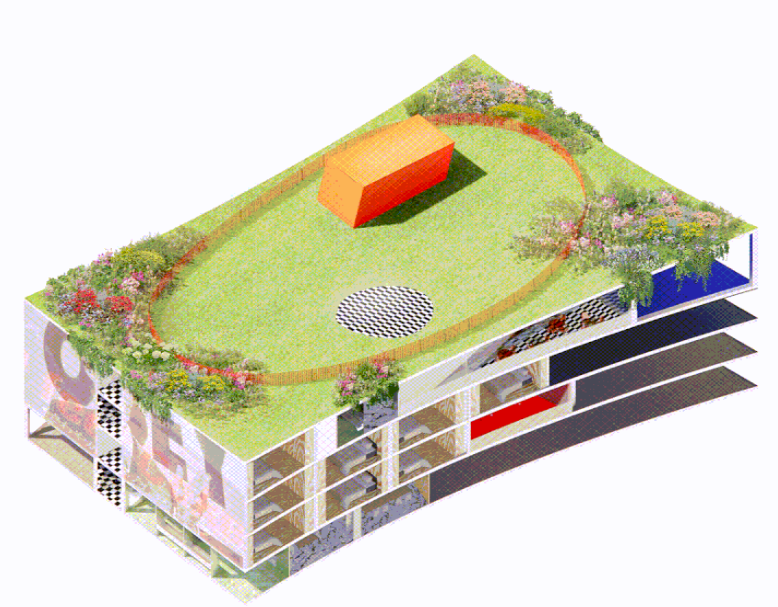
Unit-2- part-II

IoT Architecture

- IoT Architecture Components
- Comparing IoT Architectures
- A Simplified IoT Architecture
  - Core IoT functional Stack
  - Iot data management and compute stack

Comparing IoT Architectures

- In the past several years, architectural frameworks standards and have emerged to address the challenge of designing massive-scale IoT networks.
- The foundational concept in all these architectures is supporting data, process, and the functions that endpoint devices perform.
- Two of the best-known architectures
  - **oneM2M**
  - **IoT World Forum (IoTWF)**



Comparing IoT Architectures

The oneM2M IoT Standardized Architecture

- **European Telecommunications Standards Institute (ETSI)** created the M2M Technical Committee in 2008.

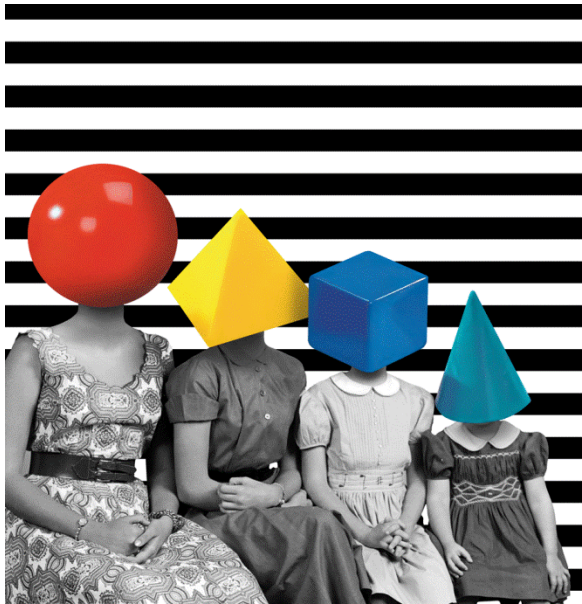
- The goal of this committee is to
  - Create a common architecture that would help accelerate the adoption of M2M applications and devices.

Comparing IoT Architectures

- In 2012 launched oneM2M as a global initiative designed to promote efficient M2M communication systems and IoT.

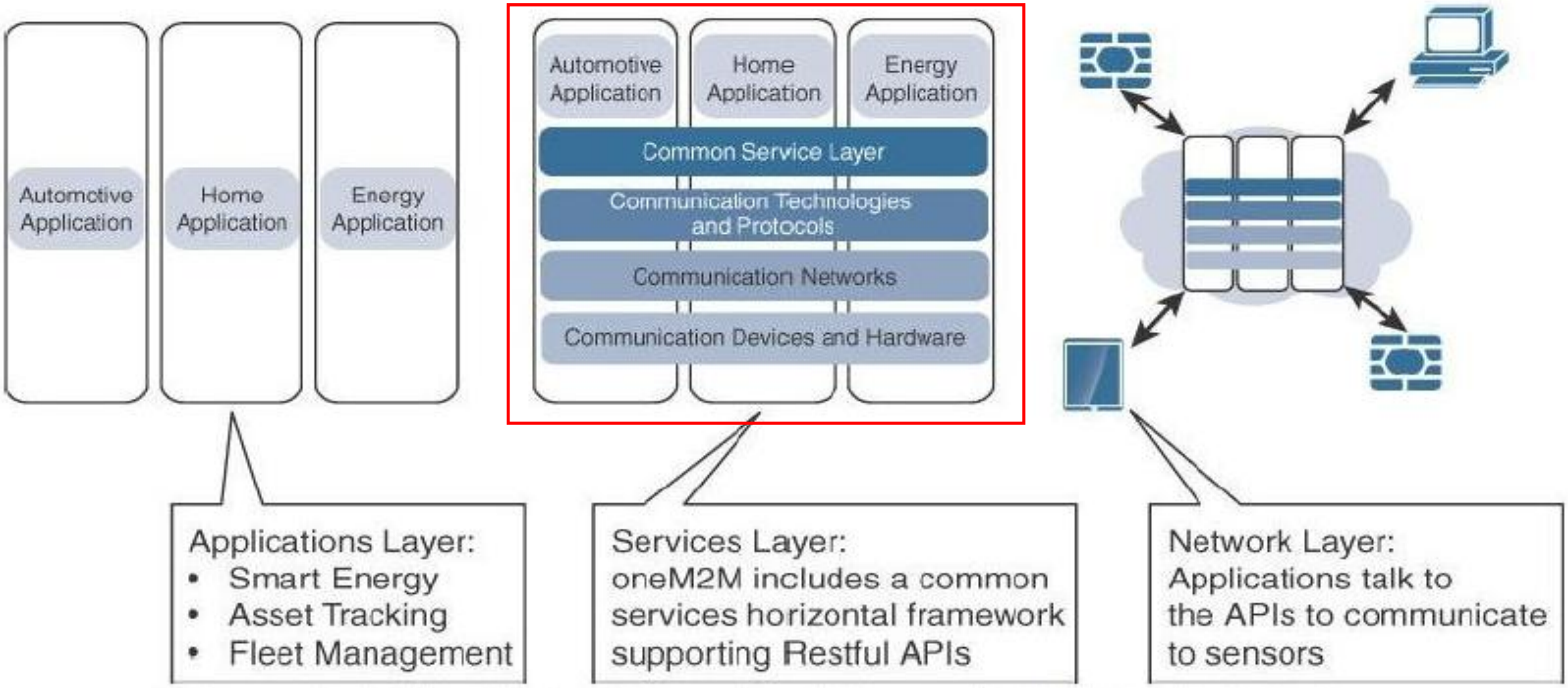
- One of the greatest **challenges** in designing an IoT architecture is dealing with the heterogeneity of devices, software, and access methods.

Heterogeneity Ex:
Shape
Colour
Size…



Comparing IoT Architectures

The Main Elements of the one M2M Architecture



Comparing IoT Architectures

The Main Elements of the one M2M Architecture

- The oneM2M architecture divides IoT functions into three major domains:

1. **application layer**,
2. **services layer**, and
3. **network layer**.

Comparing IoT Architectures

The Main Elements of the one M2M Architecture

i. Applications layer

- The oneM2M architecture gives major attention to connectivity between devices and their applications.
- This domain includes the application-layer protocols and attempts to standardize API with definitions for interaction business intelligence (BI) systems.

Comparing IoT Architectures

The Main Elements of the one M2M Architecture

ii. Services layer

- This layer is shown as a horizontal framework across the vertical industry applications.

- At this layer, horizontal modules include the physical network that the IoT applications run on, the underlying management protocols, and the hardware.

Comparing IoT Architectures

The Main Elements of the one M2M Architecture

iii. Network Layer

- This is the communication domain for the IoT devices and endpoints.
- It includes the devices themselves and the communications network that links them.
- the communications infrastructure include wireless mesh technologies, such as IEEE 802.15.4, and wireless point-to-multipoint systems, such as IEEE 801.11
- Also included are wired device connections, such as IEEE 1901 power line communications.

Comparing IoT Architectures

The IoT World Forum (IoTWF) Standardized Architecture

- In 2014 the IoTWF architectural committee (led by Cisco, IBM, Rockwell Automation, and others) published a seven-layer IoT architectural reference model.
- visualizing IoT from a technical perspective.
- Each of the seven layers is broken down into specific functions, and security encompasses the entire model.

Comparing IoT Architectures

IoT Reference Model Published by the IoT World Forum

Levels

7 **Collaboration & Processes**
(Involving People & Business Processes)

6 **Application**
(Reporting, Analytics, Control)

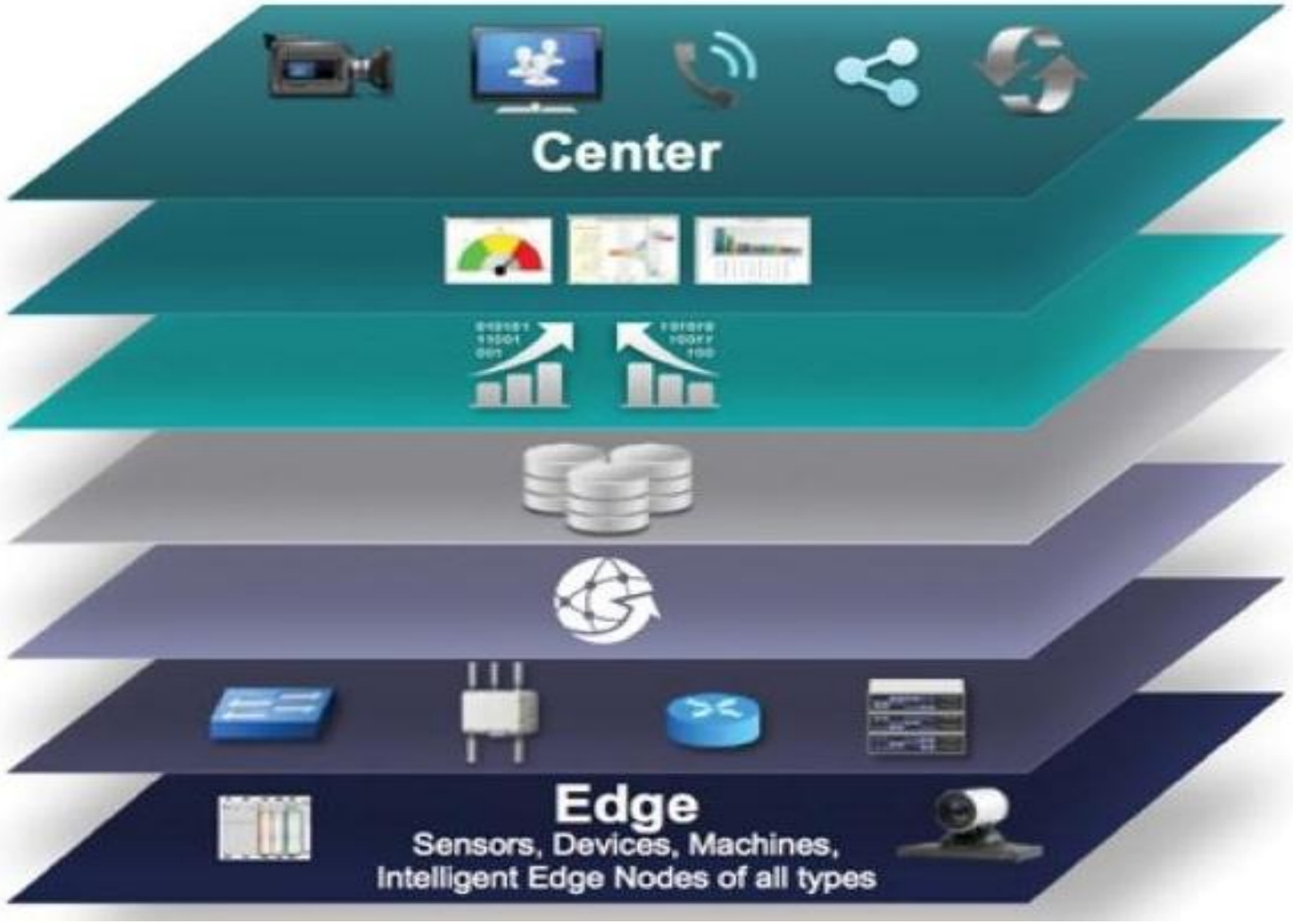
5 **Data Abstraction**
(Aggregation & Access)

4 **Data Accumulation**
(Storage)

3 **Edge Computing**
(Data Element Analysis & Transformation)

2 **Connectivity**
(Communication & Processing Units)

1 **Physical Devices & Controllers**
(The "Things" in IoT)



Comparing IoT Architectures

IoT Reference Model Published by the IoT World Forum

- Using this reference model, we are able to achieve the following:

1. Decompose the IoT problem into smaller parts
2. Identify different technologies at each layer and how they relate to one another
3. Define a system in which different parts can be provided by different vendors
4. Have a process of defining interfaces that leads to interoperability
5. Define a tiered security model that is enforced at the transition points between levels

Comparing IoT Architectures

IoT Reference Model Published by the IoT World Forum

- **Layer 1: Physical Devices and Controllers Layer**

- This layer is home to the "things" in the Internet of Things
  - including the various endpoint devices and sensors that send and receive information.

- The size of these "things" can range from almost **microscopic sensors** to **giant machines** in a factory.

- Their **primary function** is generating **data** and being capable of being **queried and/or controlled over a network.**

Comparing IoT Architectures

IoT Reference Model Published by the IoT World Forum

- **Layer 2: Connectivity Layer**
  - the focus is on connectivity.
  - The most important **function** of this IoT layer is the **reliable and timely transmission of data**.

Comparing IoT Architectures

IoT Reference Model Published by the IoT World Forum

- **Layer 2: Connectivity Layer**

IoT Reference Model Connectivity Layer Functions

② **Connectivity**
(Communication and Processing Units)

Layer 2 Functions:

- Communications Between Layer 1 Devices
- Reliable Delivery of Information Across the Network
- Switching and Routing
- Translation Between Protocols
- Network Level Security



Comparing IoT Architectures

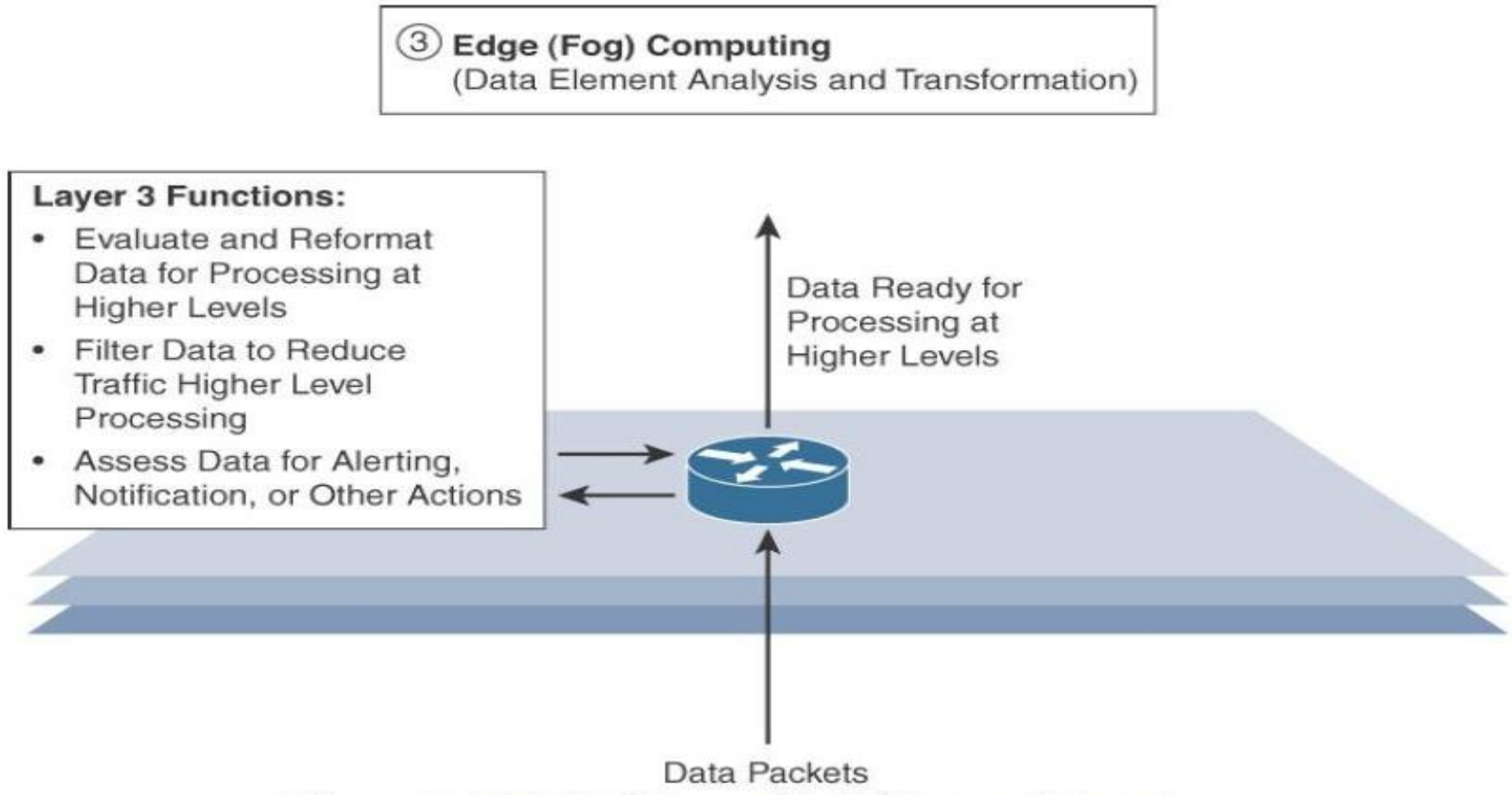
IoT Reference Model Published by the IoT World Forum

- **Layer 3: Edge Computing**
  - Edge computing is often referred to as the “fog” layer
  - At this layer, the emphasis is on **data reduction and converting network data flows into information that is ready for storage and processing by higher layers**.
  - One of the basic principles of this reference model is that **information processing** is initiated as early and as close to the edge of the network as possible

Comparing IoT Architectures

IoT Reference Model Published by the IoT World Forum

IoT Reference Model Layer3 Functions



Comparing IoT Architectures

IoT Reference Model Published by the IoT World Forum

- **Upper Layers: Layers 4–7**
  - The upper layers deal with handling and processing the IoT data generated by the bottom layer.

IoT Reference Model Layer	Functions
Layer 4 : Data Accumulation Layer	Captures data and stores it so it usable by applications when necessary. Converts event-based data to query based processing.
Layer 5: Data Abstraction Layer	Reconciles multiple data formats and ensures consistent semantics from various sources. Confirms that the data set is complete and consolidates data into one place or multiple data stores using visualization.

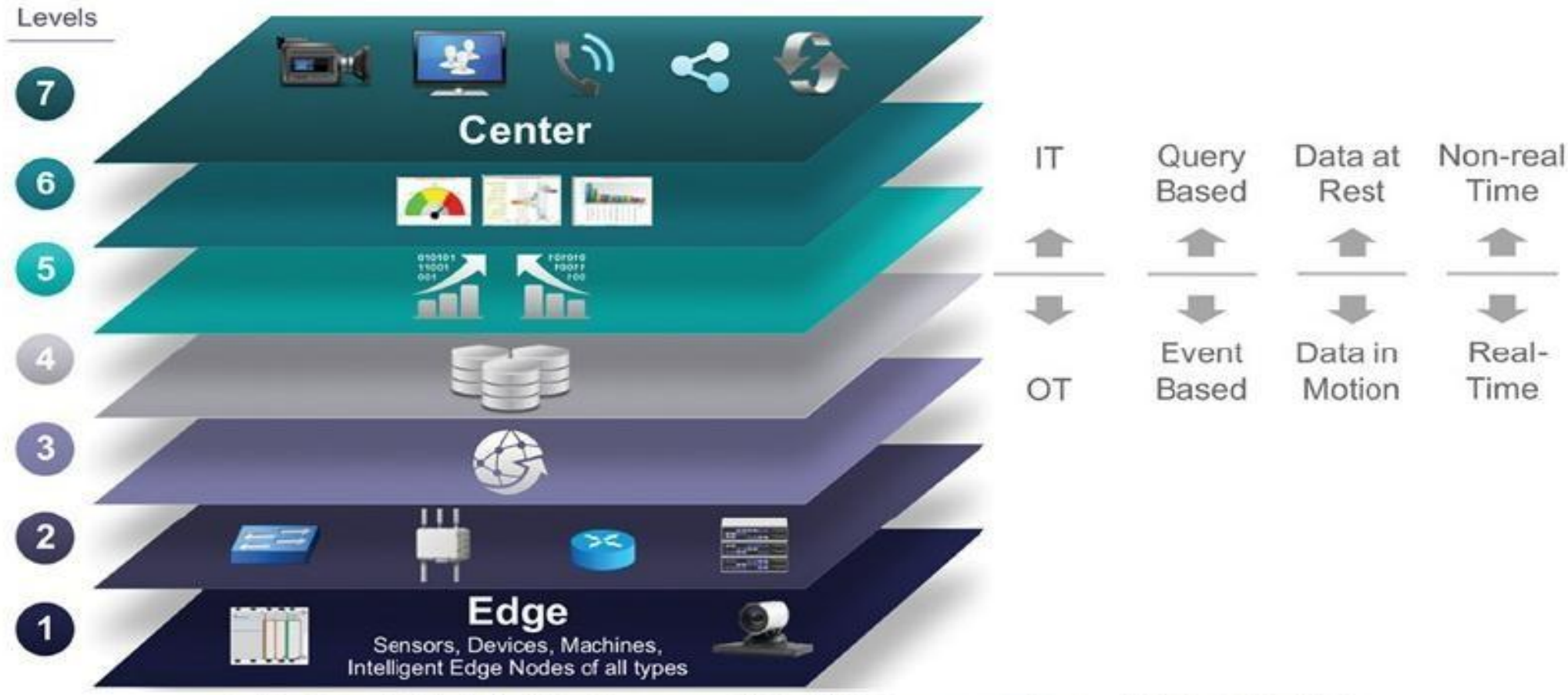
Comparing IoT Architectures

IoT Reference Model Published by the IoT World Forum

Layer 6: Applications Layer	Interprets data using software applications. Applications may monitor, control and provide reports based on the analysis of the data.
Layer 7: Collaboration and processes layer	Consumes and shares the application information. Collaboration on and communicating IoT information often requires multiple steps and it is what makes IoT useful. This layer can change business processes and delivers benefits of IoT

Comparing IoT Architectures

- **IT and IOT Responsibilities in the IoT Reference Model**
  - **IoT Reference Model Separation of IT and IoT**



Comparing IoT Architectures

- **IT and IOT Responsibilities in the IoT Reference Model**
  - **IoT Reference Model Separation of IT and IoT**
  - The bottom of the stack is generally in the domain of IOT.
  - For an industry like oil and gas, this includes sensors and devices connected to pipelines, oil rigs, refinery machinery, and so on.
  - The top of the stack is in the IT area and includes things like the servers, databases, and applications, all of which run on a part of the network controlled by IT.

Comparing IoT Architectures

- **IT and IOT Responsibilities in the IoT Reference Model**
  - **IoT Reference Model Separation of IT and IoT**
  - At the bottom, in the IOT layers, the devices generate real-time data at their own rate
  - data has to be buffered or stored at certain points within the IoT stack.
  - Layering data management in this way throughout the stack helps the top four layers handle data at their own speed.

Comparing IoT Architectures

- **IT and IOT Responsibilities in the IoT Reference Model**
  - **IoT Reference Model Separation of IT and IoT**

- the real-time "**data in motion**" close to the edge has to be organized and stored so that it becomes "**data at rest**" for the applications in the IT tiers.

- The IT and IOT organizations need to work together for overall data management.

Additional IoT Reference Models

1. Purdue model for control Hierarchy

   The Purdue Model is a hierarchical model for industrial control systems (ICS) that defines the layers of a network and their security considerations

2. Industrial Internet Reference Architecture(IIRA) by Industrial Internet Consortium(IIC)

   is a comprehensive framework developed by the **Industrial Internet Consortium (IIC)** to guide the design, implementation, and deployment of industrial internet systems (also known as the Industrial Internet of Things, or IIoT).

3. Internet of Things-Architecture(IoT-A)

   The **Internet of Things Architecture (IoT-A)** is a reference architecture framework developed to provide a unified structure for building IoT systems. It was initially part of a European research project aimed at defining an open, standardized architecture for the Internet of Things (IoT).

Unit-2- part-II

IoT Architecture

- IoT Architecture Components
- Comparing IoT Architectures
- **A Simplified IoT Architecture**
  - **Core IoT functional Stack**
  - **Iot data management and compute stack**

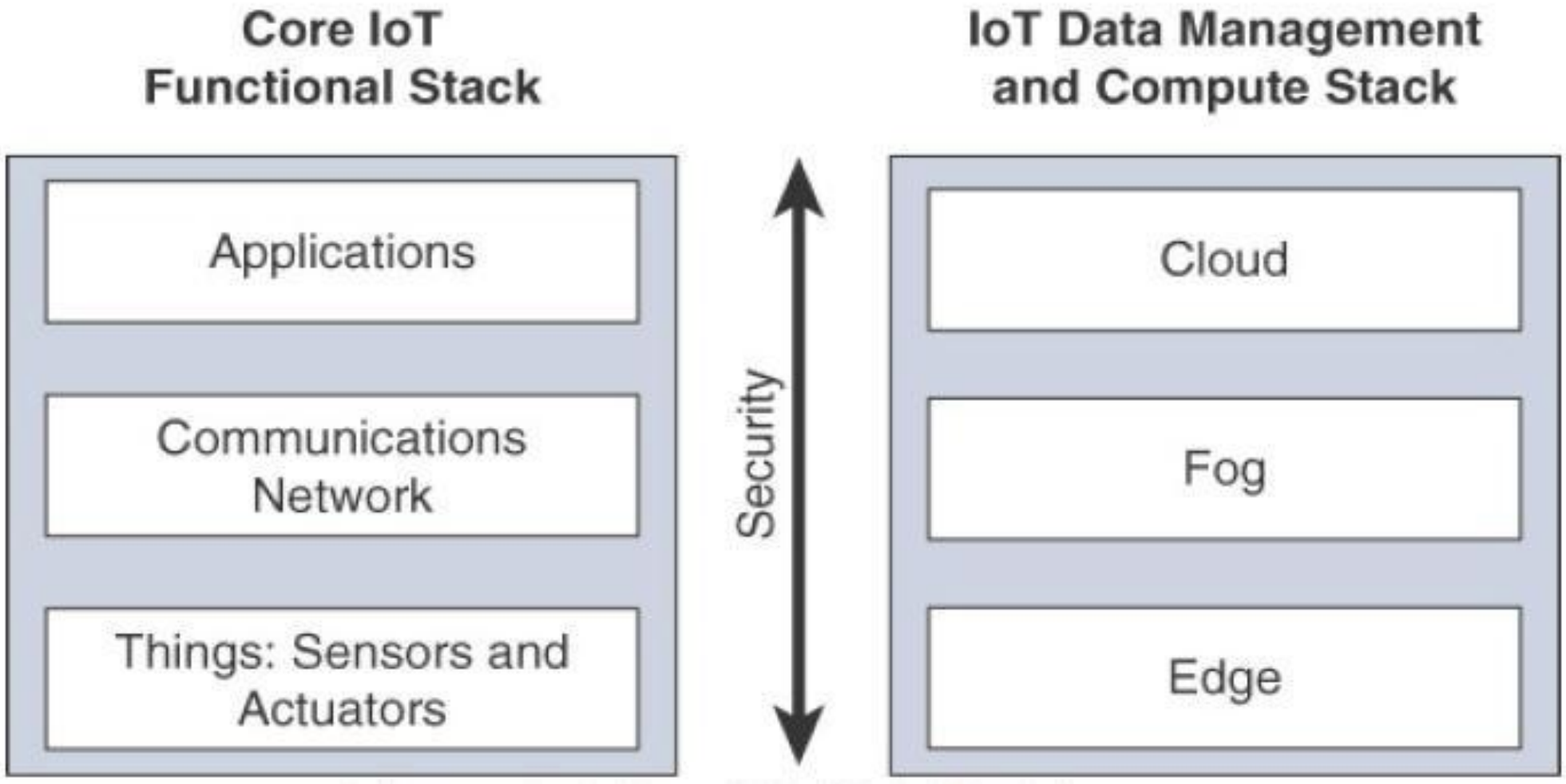
A Simplified IoT Architecture

- we can describe an IoT framework that highlights the fundamental building blocks that are common to most IoT systems and which is intended to help you in designing an IoT network.

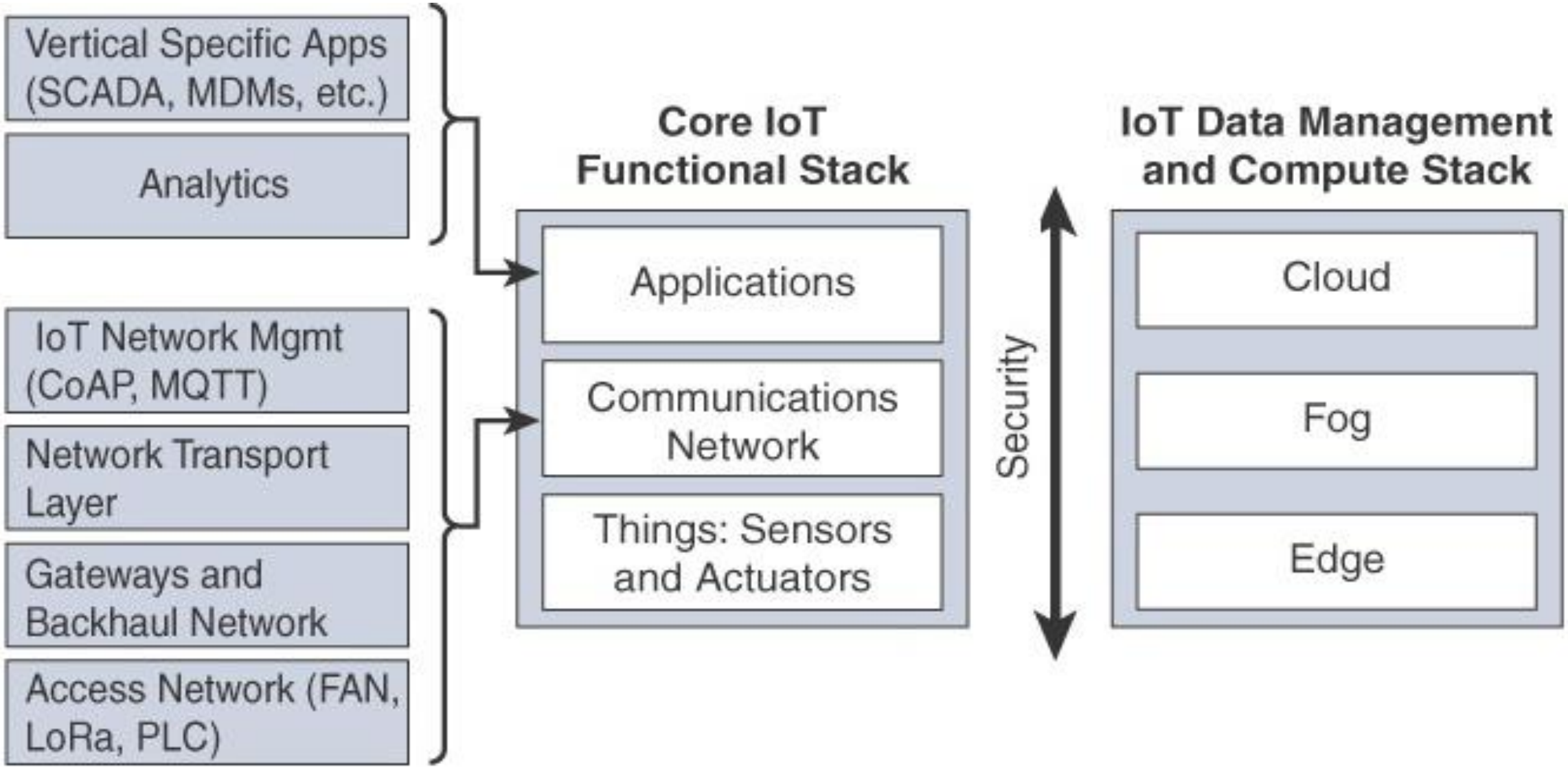
- This framework is presented as two parallel stacks:

  i. **The Core IoT Functional Stack**
  ii. **The IoT Data Management and Compute Stack**

A Simplified IoT Architecture



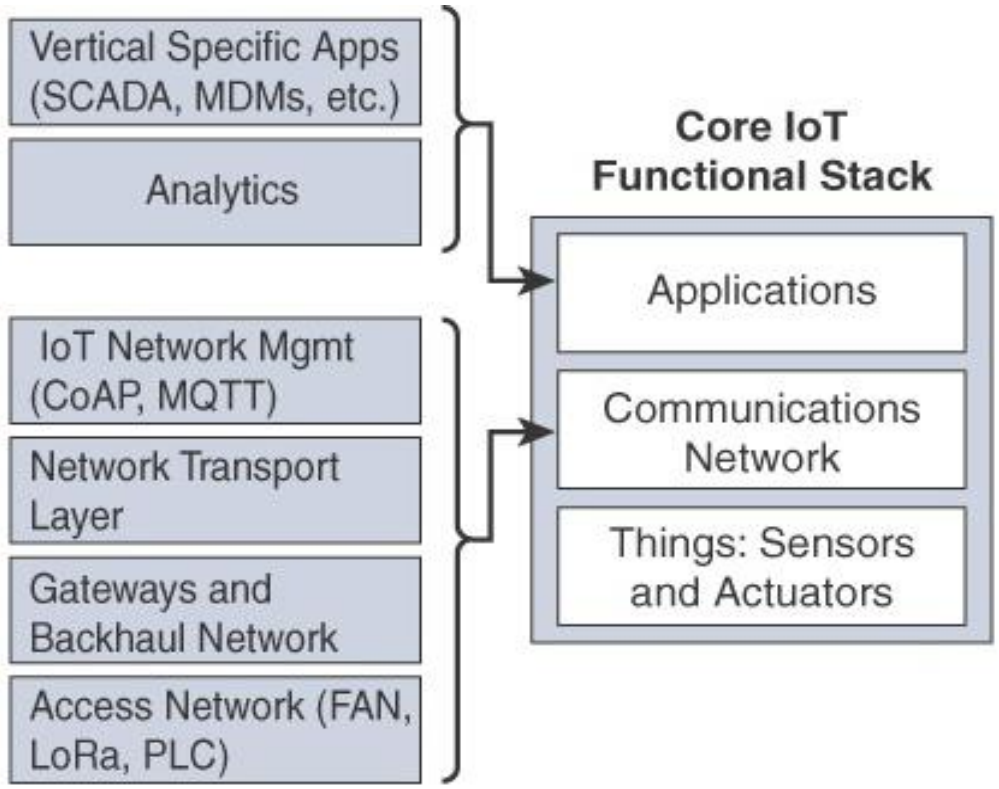
Expanded View of the Simplified IoT Architecture



The Core IoT Functional Stack

From an architectural standpoint, several components have to work together for an IoT network to be operational

- **"Things" layer**
- **Communications network layer**
  - **Access network sublayer**
  - **Gateways and backhaul network sublayer**
  - **Network transport sublayer**
  - **IoT network management sublayer**
- **Application and analytics layer**



LAYER 1

Things: Sensors and Actuators

Most IoT networks start from the object, or “thing,” that needs to be connected.

These objects are “smart” because they use contextual information and configured goals to perform actions.

A “thing” interacts with an external system to report data, exchange it with other objects, or interact with a management platform.



Variety Drives Design

The variety of smart-object types, shapes, and needs drives the variety of IoT protocols and architectures.

LAYER 1

Six Ways to Classify a Smart Object

1

Power
Battery-powered or line-powered

2

Mobility
Mobile or static placement

3

Frequency
Low or high reporting frequency

4

Data Richness
Simple or rich data per report

5

Report Range
Distance to the gateway

6

Density
Objects per cell, per gateway

LAYER 2

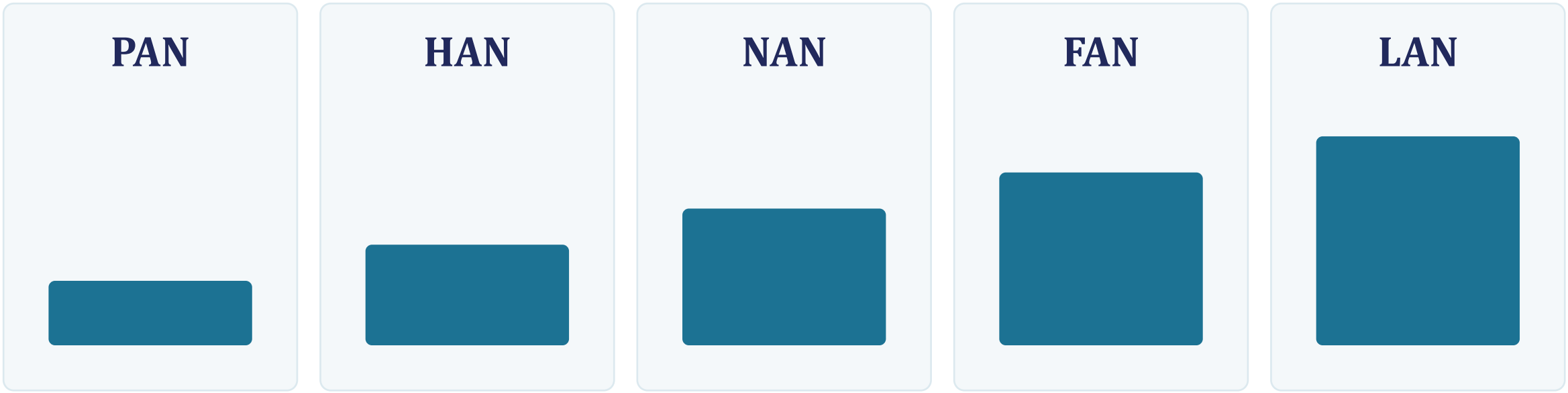
Communications Network Layer

When smart objects aren't self-contained, they must communicate with an external system — often over wireless. This layer has four sublayers:

I	**Access Network Sublayer**	How the thing reaches its gateway
II	**Gateways & Backhaul Sublayer**	Bridging to the central station
III	**Network Transport Sublayer**	TCP / UDP — reliability vs. speed
IV	**IoT Network Management Sublayer**	Lifecycle, security, optimization

Access Network: Range Categories

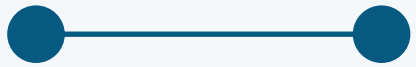
Range between the smart object and the information collector drives the choice of access technology.



A few meters	**Tens of meters**	**Hundreds of meters**	**Tens–hundreds of meters**	**Up to ~100 m**
Bluetooth	ZigBee, BLE	Group of homes	Outdoor field area	Ethernet, 802.11

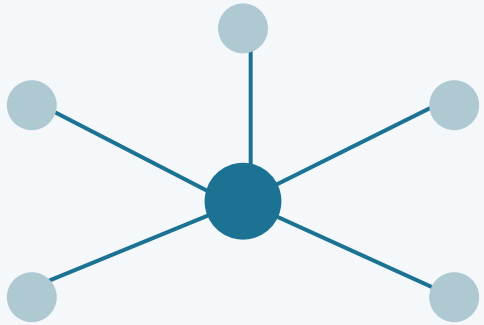
Communication Topologies

Point-to-Point



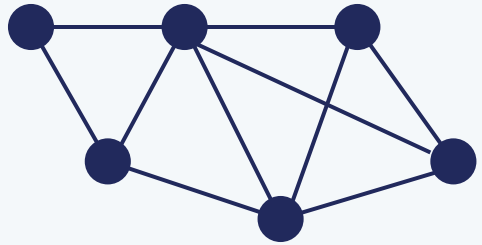
One object, one gateway — each smart object opens its own individual session.

Star / Clustered Star



Full-function devices (FFDs) coordinate reduced-function devices (RFDs) around them.

Mesh



Nodes relay through neighbors when two devices are too far apart to talk directly.

LAYER 2 · SUBLAYER II

Gateways and Backhaul Sublayer

Data collected from a smart object often needs to be forwarded to a central station in a different location.

The gateway handles this inter-medium communication — moving data from the access technology onto the backhaul.

The smart objects may be static or mobile within a limited area; the gateway itself is usually static.

Ethernet Backhaul

Used when the site is stable and locally controlled — e.g. a factory or oil & gas field.

WiMAX (802.16)

A longer-range option — up to 50 km, with data rates up to 70 Mbps.

LAYER 2 · SUBLAYER III

Network Transport: TCP vs. UDP

Responsible for end-to-end data transfer — reliability, flow control, and error handling.

TCP

Connection-oriented · guarantees delivery & order

- Establishes a connection first
- Segments data with acknowledgments & retransmissions
- Best for firmware updates, financial transactions

UDP

Connectionless · prioritizes speed over reliability

- No connection setup or delivery guarantee
- Lightweight — ideal for constrained devices
- Best for real-time sensor data, video streaming

LAYER 2 · SUBLAYER IV

IoT Network Management Sublayer

Oversees and controls the network of connected devices so they operate smoothly and securely.

Device Lifecycle

Provisioning, configuration, monitoring, firmware updates, decommissioning

Security & Access

Authentication, authorization, key management, anomaly detection

Network Optimization

Quality of Service, load balancing, fault tolerance & recovery

CoAP

HTTP-like methods (Get, Post, Put, Delete) with a shorter header. Runs on UDP.

MQTT

Broker-based: publishers, subscribers, and a broker relay messages. Runs on TCP.

LAYER 3

Applications and Analytics Layer

The topmost layer — where collected data is processed, analyzed, and turned into business or user value.

Applications

- User Interface — dashboards, mobile apps, portals
- Business Logic — the rules that drive the system
- Device Control — commands sent to lower layers

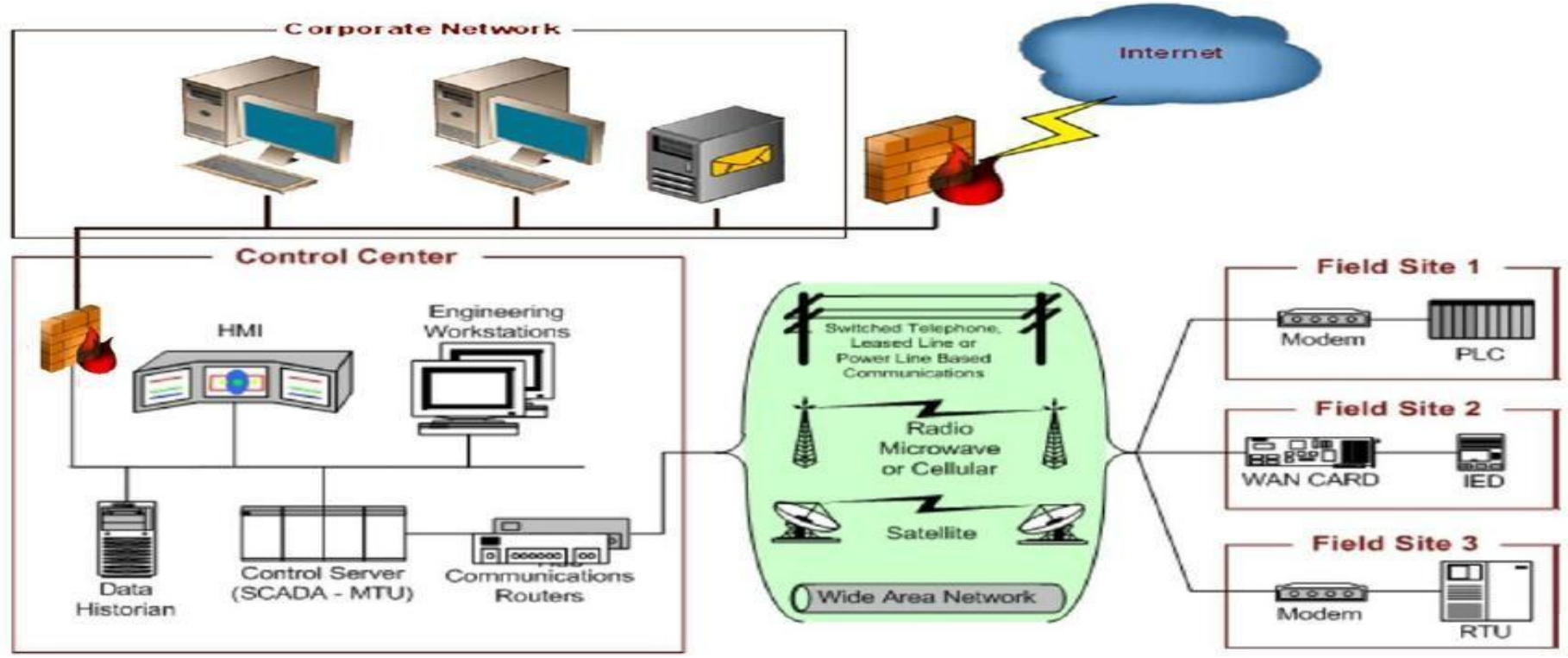
Analytics

- Descriptive — what has happened
- Predictive — what is likely to happen next
- Prescriptive — what action to take

The Core IoT Functional Stack

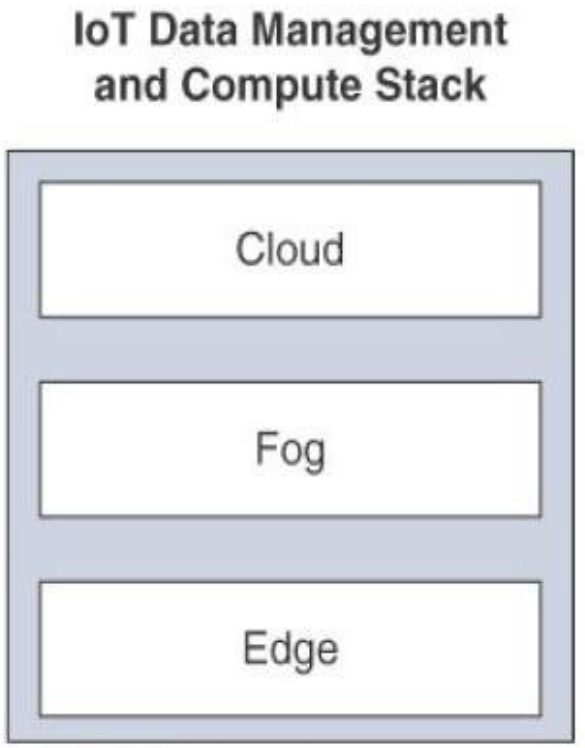
Layer 3: Applications and Analytics Layer

Ex: Supervisory Control and Data Acquisition (SCADA)



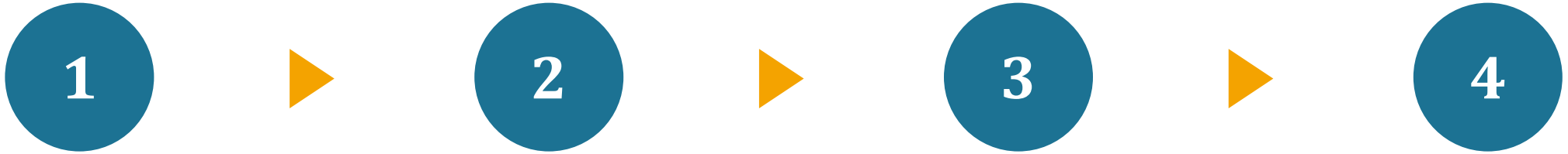
IoT Data Management and Compute Stack

- It is responsible for handling, processing, and storing the immense volume of data generated by connected devices.
- It's the engine that transforms raw data into a structured format, ready for analysis and application.



FOLLOWING ONE READING

The IoT Data Lifecycle



The IoT Data Lifecycle

From Device to Decision

1 Ingest

Data enters the pipeline



Devices / Sensors generate data

Example Tools / Platforms

- MMQTT
- AWS IoT Core
- Azure IoT Hub
- ThingsBoard
- EMQX

2 Store

Held in databases or lakes



Data is stored safely for future use

Example Tools / Platforms

- InfluxDB
- MongoDB
- MySQL
- Amazon S3
- AWS Timestream
- Azure Data Lake

3 Process

Cleaned, joined, transformed



Data is processed and prepared

Example Tools / Platforms

- Apache Kafka
- Apache Spark
- Apache Flink
- AWS Lambda
- Azure Stream Analytics
- Node-RED

4 Analyze

Patterns and decisions emerge



Insights are generated for action

Example Tools / Platforms

- Python
- Jupyter Notebook
- Power BI
- Grafana
- TensorFlow
- AWS IoT Analytics

Example: Smart Agriculture System



Soil Moisture Sensor (ESP32)

→



Ingest Data via MQTT / AWS IoT Core

→



Store in InfluxDB

→



Process & Clean Data (e.g., with Apache Spark)

→



Analyze & Visualize (Python + Grafana)

Outcome

Moisture Trends, Irrigation Alerts & Better Decisions

CHOOSING WHERE TO COMPUTE

Edge, Fog, and Cloud Compared

Edge

- Runs on the device itself
- Lowest latency, no round trip
- Limited memory and power

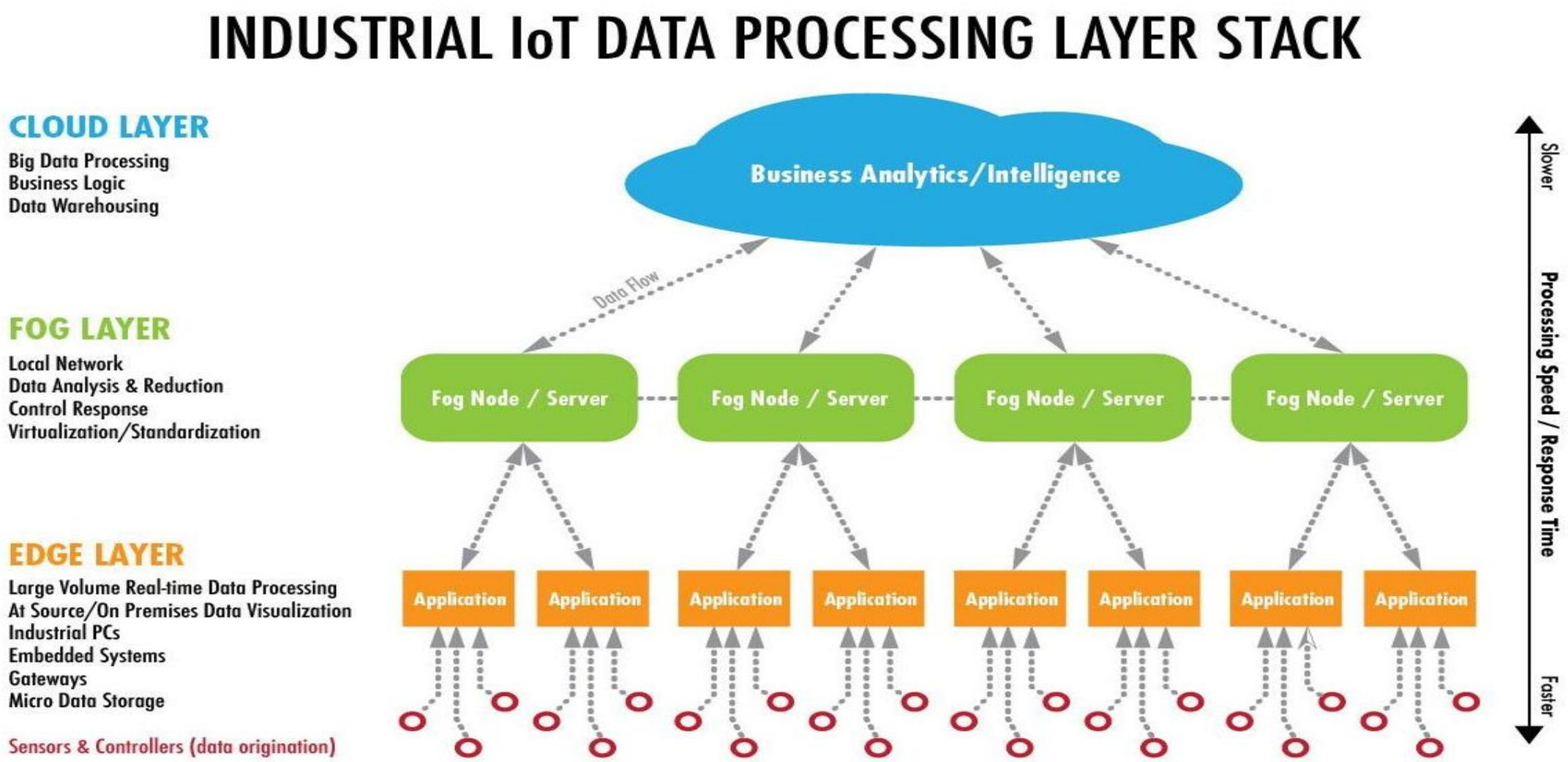
Fog

- Runs on a local gateway
- Aggregates many nearby devices
- Balances speed and capacity

Cloud

- Runs in a remote data center
- Massive storage and compute
- Best for training and history

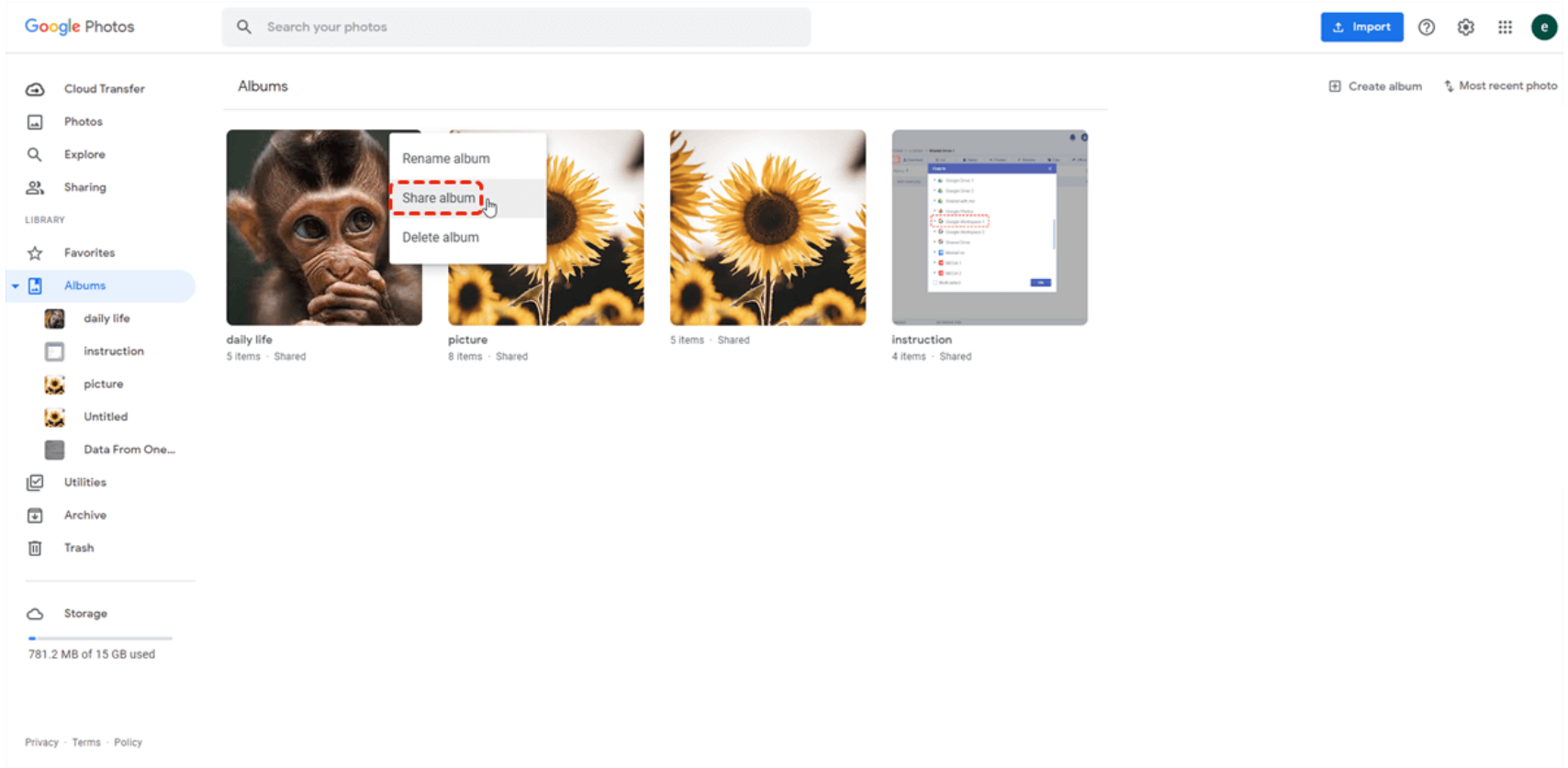
IoT Data Management and Compute Stack



IoT Data Management and Compute Stack

- Differences between Edge, Fog & Cloud Computing

Cloud Computing Example: An online photo album service where all photos you upload are sent to a central server in the cloud for storage and processing.



IoT Data Management and Compute Stack

- Differences between Edge, Fog & Cloud Computing

Edge Computing Example: A security camera that uses on-device AI to detect a person and immediately sends an alert without streaming the video to the cloud.



IoT Data Management and Compute Stack

Differences between Edge, Fog & Cloud Computing

- **Fog Computing Example**: A smart city traffic management system.
- Data from traffic cameras (edge devices) is sent to a local gateway (fog node) that processes the data to control traffic lights in real-time.
- Only summaries of traffic flow are sent to the central cloud for long-term planning and analysis.

